More Documented, Less Known: Formalization, Professional Judgment, and AI Generated Notes

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Abstract

Written records are essential to how organizations coordinate work over time, but individuals find them burdensome to both produce and use. Organizations have therefore long invested in technology, from the dictaphone to now Generative AI, to alleviate it. Drawing on two years of ethnographic research in a large health system that included the implementation of an AI scribe midway through the study, I examine how efforts to ease the documentation burden reshaped the organizational value of clinical notes. I find that records serve two distinct functions: they enable evaluation of prior work and support the continuation of subsequent work. These functions generate unequal feedback paths. Deficiencies that compromise evaluative use for billers, auditors, patients, and other audiences are returned rapidly to note authors for correction. By contrast, deficiencies that compromise subsequent clinical work are typically absorbed by the next clinician through additional searching, verification, and communication. This asymmetry produces differential coupling: records become tightly coupled to the evaluation of work but loosely coupled to the conduct of work. The AI scribe intensified this pattern by shifting clinicians from authors to ratifiers. Although it reduced the effort required to produce an evaluatively adequate note, it weakened a routine through which clinicians determined what mattered and carried those judgments forward. I theorize this dynamic as burden–value entanglement: when valuable work is embedded in a burdensome task, automation can reduce burden while inadvertently diminishing value. The findings show how technologies designed to automate record production may improve one organizational use of records while degrading another, with implications for the automation and governance of recordkeeping across organizational settings.

"I write entirely to find out what I'm thinking, what I'm looking at, what I see and what it means."

— Joan Didion

1 Introduction

Writing things down is one of the oldest technologies organizations have for governing themselves. In Weber's description of bureaucracy, the file, rather than the individual person, carries an organization's memory. Preserved records let an organization operate through roles rather than through personal ties such that official knowledge is able to outlive any single actor (Weber, 1978). Nearly every large organization still depends on this basic idea.

Yet documentation remains one of the most disliked parts of organizational life. Workers who maintain records often describe the work as burdensome, duplicative, and detached from the work they understand themselves to be doing. Doctors experience documentation as a source of burnout; salespeople resent customer databases; engineers, maintenance workers, and administrators complain that the system meant to preserve organizational knowledge instead pulls them away from "real work" (Markus, 1983; Lapointe and Rivard, 2005; Maslach et al., 2001; Herd and Moynihan, 2018). The complaint would matter less if records were peripheral. They are not. Records are where organizations preserve warnings, justify decisions, assign responsibility, coordinate handoffs, and reconstruct what happened after the fact. When records do not carry forward what future actors need to know, the consequences can be serious.

Organizations have responded to this tension largely through technology. They install new information systems, add templates and structured fields, automate data entry, redesign interfaces, and now increasingly use generative AI to draft records. Research has often followed these technologies, asking how systems are adopted, resisted, enacted, or incorporated into everyday practice (Barley, 1986; Orlikowski, 1992, 1996; Edmondson et al., 2001; Anthony, 2021). This work has shown that technology does not determine use and that use is not the same as effective use (Burton-Jones and Straub, 2006; Burton-Jones and Grange, 2013). But the persistence of the documentation problem suggests a deeper puzzle. Record-producing technologies have become faster, more integrated, more searchable, and more standardized. From the organization's vantage point, records can look better: more complete, more structured, more auditable, and more available. Why, then, do the people who create and depend on those records often experience documentation as increasingly burdensome, less useful, and less connected to the work itself?

I study this question through two years of ethnographic fieldwork inside a large academic hospital system, following the everyday work of writing, reading, correcting, and working around clinical encounter notes. Clinical notes offered a revealing setting because they were mandatory, consequential, and multi-purpose. They supported billing, audit, legal accountability, patient access, and organizational oversight. They were also supposed to help doctors and other clinicians understand what had happened to a patient and what should happen next. The same professionals who wrote notes also depended on notes written by others. This made the setting especially useful for examining how a record could remain organizationally necessary while becoming less useful for carrying forward everyday clinical work.

Because a decline in clinical usefulness is hard to detect from completed records alone, I use qualitative fieldwork to observe record production and use in practice. A note can be signed, billed upon, and available in the system while still failing to help a future clinician understand what happened and why. Those failures appeared not as noncompliance or nonuse, but in the compensatory work clinicians performed around the record: searching, skipping, verifying, calling, remembering, rewriting, and asking patients to reconstruct what the note did not carry. Fieldwork allowed me to follow these repairs in real time and ask whether the failure changed the record or changed the work clinicians did around it.

I find that the clinical note was pulled in two directions. When the note failed formal audiences such as billing, compliance, audit, or patients, the failure tended to return quickly to identifiable clinician-authors through queries, correction requests, training, interface changes, or workflow restrictions. When the note failed clinicians as readers, however, the failure was usually repaired locally. Clinicians searched elsewhere in the chart, asked patients, called colleagues, relied on memory, or used messages and handoffs to recover what the note did not provide. These repairs protected immediate care, but they rarely returned as feedback on the note itself. I call this process *asymmetric correction*: deficiencies in accountability value were visible, attributable, and enforceable, so they changed the record; deficiencies in clinical usefulness were compensated for locally and resulted in behavior change.

Over time, this produced a second process I call *recursive withdrawal*. As clinicians found notes less useful, they relied on them less. As they relied on notes less, they had less reason to expect others to read what they wrote. And as anticipated readership declined, careful clinical authorship became less rewarding. The note did not disappear. It remained mandatory, signed, circulated, billed upon, audited, and available for review. But the relationship between professional reading and professional writing weakened. When the note failed a formal audience, the record changed; when it failed a clinical reader, the work changed.

Midway through the fieldwork, the hospital introduced an ambient AI scribe that drafted encounter notes from recorded clinician-patient conversations. The tool produced real and immediate benefits: clinicians finished notes faster, reduced after-hours documentation, and consultant recommendations were implemented earlier, reducing delays in care. But by making notes that met evaluative standards cheap to produce, it changed clinicians' relation to the record. Clinicians increasingly moved from authoring an account to ratifying a generated draft. In doing so, the AI scribe relieved a genuine burden, but it also weakened one of the remaining practices through which clinical judgment entered the note: the act of processing and distilling one's observations into an account for future use.

The AI scribe did not create the divergence between a record's ability to satisfy evaluative standards and its ability to help carry clinical work forward. It entered a record system already shaped by asymmetric correction and recursive withdrawal. But it did make the pattern more visible. I call this a *burden-value entanglement*: a condition in which the same practice imposes unwanted burden while also producing a distinct form of value. The mechanism is that organizations can remove the visible cost of producing a record before they have rebuilt the less visible process through which judgment enters it. The value was not in the burden itself, but in the authorial work still bundled with that burden: deciding what mattered, distilling the encounter, and leaving a usable account for someone who was not there.

This study makes three contributions. First, it brings documentation itself into organizational theory as work, not merely as evidence of work. Records become useful not just by existing. Their value is produced through the linked practices of writing and reading. Second, it explains how a single mandatory record can become differentially coupled: increasingly responsive to accountability standards and decreasingly useful for practice and coordination. This extends theories of decoupling by showing how unequal feedback can improve one function of a record while eroding another (Meyer and Rowan, 1977; Bromley and Powell, 2012; Pache and Santos, 2013). Third, it clarifies what is at stake when organizations automate documentation. The danger is that some of what appears to be burden may also be the work through which judgment, memory, and coordination enter the record. Technologies that remove the burden without rebuilding that value-producing function may help organizations prove that work occurred while making it harder for future workers to know what happened.

2 Theoretical Background

2.1 Records and the promise of formal organization

Written records let organizations preserve, evaluate, and coordinate work beyond the individual employees who performed and witnessed it as it occurred. In Weber's account, this is what separates the office from, say, a family, that utilizes only oral tradition. Preserved files stabilize official knowledge beyond individual officeholders and let their authority operate through roles rather than personal ties (Weber, 1978). Records solve a basic administrative problem by allowing leaders to govern work they cannot continuously observe, and in many cases cannot directly assess (Yates and Chandler, 1951; Stinchcombe, 2001). Research on computer-supported cooperative work similarly treats shared documentation as the basic infrastructure for such complex interdependent work (Schmidt and Bannon, 1992). In both accounts, the record derives its organizational value from allowing work to travel.

Travel, however, requires translation. To move beyond the setting in which it occurred, work must be selected, categorized, and rendered in a form that others can interpret. This makes records portable, but also selective: some features of work become visible and comparable, while others remain dependent on context. The promise of formal organization rests not only on producing records, but on sustaining a common representation across heterogeneous uses.

2.2 Multiple uses of a shared record

Research on data quality provides one way to understand this problem. Early approaches often treated quality as correspondence between recorded data and an underlying state of the world. Wang and Strong broadened this view by defining quality in relation to use: in addition to accurate, data must be relevant, interpretable, accessible, and appropriate to the task at hand (Wang and

Strong, 1996). Information systems research makes a related distinction between system use and effective use: a system may be used frequently while failing to help users accomplish the purposes for which the system matters (Burton-Jones and Straub, 2006; Burton-Jones and Grange, 2013). A record can therefore be accurate but unhelpful, available but unused effectively, or well suited to one task and poorly suited to another.

In organizations, however, neither the user nor the task is singular. Shared records are produced in one setting and consulted in many others by people doing different work. The same person may also return to a record for a different purpose. The problem is therefore how one artifact remains workable across heterogeneous uses, not just whether or not it is accurate.

Research on boundary objects explains how one artifact can support varied uses. Boundary objects are rigid enough to maintain common structure to travel across social worlds while retaining enough flexibility to be interpreted and used locally (Star and Griesemer, 1989). Studies of records similarly show that documents can support clinical, administrative, legal, financial, and organizational tasks without meaning exactly the same thing to each audience (Berg, 1996; Berg and Bowker, 1997; Pine and Bossen, 2020). This work explains how heterogeneous groups can coordinate around a shared artifact despite differences in perspective. It is less explicit about how the artifact itself changes as those groups continue to use and revise it, or whether changes that support one use preserve its capacity to support others.

Theories of coupling provide a language for examining these possibilities without assuming in advance how the uses relate. Weick described organizations as loosely coupled systems in which elements remain responsive to one another while preserving enough distinctiveness that change in one does not necessarily create change in another (Weick, 1976). Later work treated coupling as varying in both responsiveness and distinctiveness rather than as a binary condition (Orton and Weick, 1990). Institutional theory extends this argument by explaining why formal structures can persist apart from everyday activity because they confer legitimacy or serve symbolic purposes (Bromley and Powell, 2012; Feldman and March, 1981; Meyer and Rowan, 1977). Formal representations and substantive practice may therefore remain linked while following different trajectories.

Research on quantification, audit, and organizational control clarifies why some uses acquire more force than others. Formal systems make some features of work visible and comparable while leaving others difficult to specify (Espeland and Stevens, 1998; Stinchcombe, 2001). Organizations therefore rely more heavily on formal, output-based controls when performance can be verified and on informal or relational controls when it cannot (Ouchi, 1979). By specifying what can be observed, compared, and reviewed, these systems also shape what the organization recognizes as work.

When multiple standards coexist, organizations can sometimes accommodate them without fully reconciling them. Research on selective coupling shows how organizations combine selected practices associated with competing logics rather than adopting any one logic in full (Pache and Santos, 2013). In healthcare, for example, managerial and medical logics carry different understandings of what constitutes quality work (Reay and Hinings, 2009). Research on professions has examined the related tension between administrative control and professionals' authority over the substance and standards of their work (Freidson, 2001). These accounts show how competing standards can be distributed across roles, units, and practices.

A shared mandatory record, however, is difficult to partition in this way. Different demands converge on the same artifact, and the same people who produce it may later depend on it for another task. Existing research explains why artifacts can span communities, why formal representations can depart from practice, and how organizations can accommodate competing logics. It offers less guidance on how one record changes when it must continue to serve several tasks over time. The unresolved question is whether and how changes made for one use affect the record's capacity to support others.

2.3 Technology, burden, and record system design

When records become burdensome or fail to meet organizational expectations, organizations usually try to redesign the systems through which they are produced. The standard explanation for resistance to documentation treats it as a problem of motivation or design, one that would dissolve if people complied more fully or the system asked less of them. This echoes a long history of research that links technology adoption with perceived usefulness and ease of use (Davis, 1989; Venkatesh, 2003). Yet adoption is a limited test in mandatory settings, where use can continue without acceptance and compliance can continue without effective use (Burton-Jones and Grange, 2013).

In medicine, this logic has been especially visible because documentation burden is now measured in clicks, time, and after-hours work, and is linked to clinician burnout and turnover (Arndt et al., 2017; Sinsky et al., 2016; Apathy et al., 2023). The recurring managerial wager is therefore that technology can preserve the organizational benefits of documentation while reducing the effort required to create it. Organizations therefore install, customize, replace, and automate shared record systems in the hope that records can become faster to produce, easier to standardize, and more useful.

Scholarship has therefore largely followed the technology used to produce and share records. One stream asks how new technologies disrupt routines and role relations, redistribute control, change how workers learn, and reshape the exercise of professional expertise (Barley, 1986; Edmondson et al., 2001; Beane, 2019; Kellogg et al., 2020; Anthony, 2021). A complementary applied literature has made documentation burden measurable, specifically counting time spent, clicks, and after-hours work required to maintain record systems, thereby motivating redesign and automation aimed at producing records with less effort (Apathy et al., 2023; Arndt et al., 2017; Sinsky et al., 2016).

Research on collaborative technologies and shared repositories further complicates the relationship between easier production and organizational value. Contributors may bear immediate costs while those who benefit are distant or unknown to them. Making a system available does not ensure that workers will integrate it into their daily practice, and information removed from the setting in which it was produced can be difficult to interpret and reuse elsewhere (Grudin, 1988; Markus, 2001; Orlikowski, 1992, 1996). Research on organizational learning and capability traps adds that workers often repair immediate breakdowns without changing the systems that produced them, allowing local performance to continue without interruption while the recurring problems remain hidden (Repenning and Sterman, 2002; Tucker and Edmondson, 2003).

Together, these literatures explain why record-system redesign can alter work in unexpected ways, why reducing the cost of contribution may not improve subsequent use, and why local adaptation can allow deficiencies to persist. The unresolved issue is whether making a record easier to produce preserves the several kinds of value the record is expected to carry.

2.4 Writing records and producing organizational knowledge

One reason record-system redesign may have consequences beyond burden is that producing a record is not always separate from the work the record describes. A representational view treats documentation as a proxy for a real-world state: when the system and the person entering the data are both working well, the record should correspond accurately to what occurred (Burton-Jones and Grange, 2013). From this perspective, producing the record creates an artifact whose quality and use can be assessed after it exists.

A second view treats producing the record as part of the work itself. Records make action portable, visible, and auditable, but writing itself can require professionals to select what matters, interpret what happened, and communicate judgment to others (Berg, 1996; Berg and Bowker, 1997). Like other organizational routines, record production has both an ostensive dimension (an idealized template of what a complete record contains), and a performative dimension (the situated, in-the-moment act of writing it), and the two need not stay aligned (Feldman and Pentland, 2003).

These perspectives are not mutually exclusive. A record may represent judgment while the process of producing it also contributes to the formation and articulation of that judgment. This point is especially important in expert work, where some knowledge remains tacit: actors know more than they can fully specify in advance (Polanyi, 1966). Writing does not eliminate tacitness, but it can force a partial conversion of situated judgment into explicit form. In that sense, record production is one practice through which organizations move knowledge from persons and situations into artifacts that others can use (Nelson and Winter, 1982; Walsh and Ungson, 1991).

This constitutive view has a longer history in research on writing and cognition. Writing is more than the transcription of thought already formed; it can reorganize what is classified, remembered, and understood (Goody, 1986). The physical act of writing something down, rather than simply reading or hearing it, can then change what people later remember and understand (Mueller and Oppenheimer, 2014). Sociomaterial accounts make a related point: the form of a technology and the activities organized through it cannot be treated as independent (Orlikowski and Scott, 2008). Changing how a record is produced may therefore alter when, where, and by whom interpretation occurs, in addition to the effort required and the appearance of the resulting artifact.

If writing helps constitute judgment rather than only record it, then where that judgment ends up living is not a forgone conclusion. Research on organizational memory identifies several possible repositories, including the people who do the work, the structures and roles built around it, and external archives such as records and databases, without assuming that any one of these is the natural or correct place for a given piece of knowledge to sit (Walsh and Ungson, 1991). The practices through which records are produced help determine what becomes externalized, what remains with particular people, and what must later be reconstructed through routines, relationships, or informal communication.

These perspectives give us reason to distinguish the finished record from the practice through which it is produced. Yet the implications of that distinction for record-system redesign remain unclear.

Shared records are expected both to make prior work available for evaluation and to support the work that follows. Organizations repeatedly seek to reduce the effort required to produce them, but existing research does not establish whether changing the production process leaves these functions intact, improves them together, or affects them differently. I therefore ask: *How do efforts to reduce documentation burden affect a shared record's capacity both to make prior work available for evaluation and to carry work forward?*

3 Methods

To study this question, I conducted a two-year inductive qualitative field study at Yankee Health, a major academic health system. I initially entered the field with a broad question: why had electronic documentation become so burdensome for clinicians, and how did they produce and use patient records in everyday care? A longitudinal qualitative approach was appropriate because it allowed me to get close to the phenomenon of interest and observe in fine-grained detail how patient notes were written, read, and corrected in situ. I was able to compare clinicians' explanations with their enacted practices and follow changes in those practices over time (Barley, 1990; Van Maanen, 1979). Sustained presence also helped me build the trust necessary for candid accounts of how clinicians experienced deficient records as affecting their work and patient care (Locke, 2003).

3.1 Research setting

Yankee Health comprises nearly 500 clinical sites across 185 towns and employs around 40,000 staff and 6,000 physicians. The system encompassed a wide range of care environments, including large urban hospitals, suburban clinics, and rural primary care practices, and served a demographically diverse patient population. The system had used Odyssey, a leading commercial Electronic Medical Record (EMR), since 2013. Yankee Health also possessed substantial technical, financial, and administrative capacity, and its leaders explicitly sought to improve documentation and data quality. These features reduce the likelihood that the observed difficulties arose only from the absence of basic technical infrastructure or organizational support, although they do not rule out problems of design or implementation.

The breadth of the system offered useful variation in who would later depend on a note. In inpatient medicine and many hospital-based specialties, continuity was largely interpersonal: night teams inherited patients from day teams, consultants followed the hospitalist, and clinicians routinely acted on notes written by someone else. In primary care and certain outpatient specialties, continuity was more often intrapersonal: the clinician reading an earlier note was frequently the person who had written it, whether days, months, or years before.

Eighteen months into the fieldwork, Yankee Health introduced Traverse, an ambient AI note-writing tool that generated draft encounter notes from recorded clinical conversations. Traverse was first deployed to a pilot group of outpatient primary care providers, then to the broader outpatient provider population, and later to inpatient providers. Because I was already observing both inpatient and outpatient work, the rollout provided a contemporaneous pre-introduction baseline and an opportunity to observe the early shift from composing notes to reviewing generated drafts. These data support claims about changes in note-production practice, not about the tool's long-run equilibrium or downstream clinical effects.

3.2 The clinical encounter note

The clinical note is one component of the broader patient record, which accumulates a patient's medical history, diagnoses, orders, test results, medications, procedures, and care plans. Within that record, encounter notes (progress notes, operative notes, consult notes, discharge summaries, and others) serve a distinctive role. They are the place where clinicians are expected both to document what occurred, and to render the patient's condition clinically intelligible: what is going on? What matters? What has been ruled in or out? What is the plan, and why?

Notes therefore combine narrative and structured material. Clinicians could type or dictate narrative sections, copy forward prior notes, use templates, or insert "dot phrases" that automatically populated information such as laboratory results or recorded weights. The two types of data are not cleanly separated, and where the boundary falls is largely the clinician's prerogative. Some fold lab values and vitals into their narrative, while others judge this redundant as the same data are accessible elsewhere in the record. Two notes may be formally complete and differ greatly in how much interpretation and clinical judgment they convey.

The note also served multiple audiences: clinicians used it in subsequent care, while administrators, payers, regulators, and patients evaluated it as an account of prior work. A single clinician might document more than twenty encounters in a day, sometimes with as little as one hour formally allocated for administrative work. This made the note a useful site for studying how clinicians produced a record under time pressure for audiences whose definitions of adequacy did not always align.

3.3 Data collection

Fieldwork was conducted between April 2024 and April 2026 and comprised site visits, direct observation of day-to-day clinical and administrative work, formal and ethnographic interviews with clinicians, and informal conversations. Data collection unfolded as allowed by coordination with individual providers, with breaks between time in the field used to expand fieldnotes and refine subsequent sampling. Table 1 details data sources and their uses. Figure 1 details the timing of fieldwork against the Traverse rollout.

From April through September 2024, I attended weekly meetings with senior hospital leaders to build organizational legitimacy, refine the study focus, and vet the design with key stakeholders while institutional approvals were secured. This time is indicated in Figure 1 with a different color as these foundational meetings were largely on zoom and I was not physically in the field. I treated these meetings as part of access-building and study development rather than as empirical observations: they were not coded, included in the analytic corpus, or counted among the meetings reported in Table 1. From October 2024 through April 2025 I concentrated interviews and observation on primary care physicians and hospitalists, for whom documentation burden was judged to be particularly acute. I returned from August to November 2025, and again from February through April 2026. As early fieldwork showed that clinicians' burden depended partly on records produced elsewhere in the system, I expanded observation across inpatient and outpatient specialties. When a provider mentioned another provider's notes as particularly burdensome to get through or helpful for synthesis, I endeavored to shadow the author of those notes, leading to direct understanding of the interdependent nature of clinician reliance on the record.

Observation. I conducted over 300 hours of direct observation across inpatient and outpatient settings including emergency medicine, pulmonology, nephrology, primary care, and intensive care. I shadowed attending physicians, residents, physician assistants, nurses, and patient care technicians through patient encounters, chart review, handoffs, note composition, and the interruptions and compensation surrounding those activities.

My role was primarily observational, although I occasionally assisted with mundane, non-clinical tasks such as fetching liver ultrasound machines, carrying supplies between rooms, helping patients in and out of chairs and beds, and logging out of Odyssey on several physicians' office computers so that the note could be completed on the computer in the exam room. This quasi-active role facilitated access and rapport without involving me in clinical decision-making, diagnosis, or treatment (Adler and Adler, 1987). Initial presence in clinical spaces, especially at the bedside, was often met with expected caution, but over time I became a familiar presence. Clinicians regularly invited me to huddles, resident teaching sessions, optimization meetings, and informal discussions, and on occasion into the day's improvised routines when operations were disrupted - for instance, an emergency-department-wide Taco Bell break when the power in the whole city went out (a 95-degree, humid ED is not for the faint of heart).

Observation centered on clinicians' real-time interactions with Odyssey: what they opened and ignored, which sections they trusted, how they searched for missing context, when they contacted another person, what they wrote, and how they decided a note was complete. These activities were often tacit or automatic, making direct observation necessary to compare clinicians' general accounts of documentation with their enacted practices. I also observed in-basket work, including patient messages that questioned, contested, or sought clarification of what had been documented. These exchanges allowed me to observe directly how patients evaluated the record and how clinicians responded to those evaluations.

I also observed meetings on billing and coding, Odyssey workflows, and hospital research initiatives. These meetings showed how leaders, clinicians, administrators, and IT personnel framed documentation problems and negotiated tensions among burden, compliance, and record quality.

Interviews. To complement observation and probe clinicians' own interpretations of documentation work, I conducted 57 interviews with attending physicians, residents, nurses, and administrative staff. Of the interviews, 30 were formal and were conducted privately over Zoom or in person, lasted 30–90 minutes (most typically an hour), and were recorded and transcribed. The remaining interviews were ethnographic, free-form, and took place as time allowed, such as during lunch, no-show appointments, or after the workday had ended. These conversations were correspondingly closer to the work in both time and space and more often reflected the day's particular points of friction. Most participants were interviewed once, and a small subset were followed up with as emerging insights warranted.

Interview guides were tailored by role but consistently addressed how participants understood the purpose of the electronic record, how they created and used notes, which parts of documentation supported or obstructed their work, and how they responded when the record did not provide what they needed. Where possible, I interviewed clinicians before or during periods of observation and compared what they said with what they consequently did.

The empirical materials include handwritten field notes filling more than a dozen notebooks, which I expanded into detailed fieldnotes and analytic memos after sessions in the field. Together these materials amounted to several hundred thousand words. The fieldnotes captured both what clinicians said about documentation, as well as the sequence and setting of their documentary activities: when a problem became apparent, who noticed it, how the problem traveled, who compensated, and whether the record or the supporting work changed. Across all modes, the aim was to observe how documentation was produced and used in practice and how clinicians' relationship to the record changed over time.

Table 1: Data sources

Data source	Volume	Time period	Roles/specialties covered	Primary analytic use
Direct observation	300+ hours	Sep 2024–Apr 2026	EM, pulm, nephrology, primary care, ICU	Real-time documentation practice, in-basket responses, workarounds, handoffs
Formal interviews	30 interviews, 30–90 min each	Sep 2024–Apr 2026	Attendings, residents, nurses, admin staff	Participants' own accounts of documentation purpose/use
Ethnographic/ informal interviews	27 conversations	Oct 2024–Apr 2026	Same as above	Real-time reactions to friction points
Meetings included in corpus	5	Apr 2024–Apr 2026	Billing/coding, Odyssey workflow, leadership	Organizational framing of documentation policy
Fieldnotes/ analytic memos	12+ notebooks, several hundred thousand words	Ongoing	—	Process tracing, causal loop diagrams

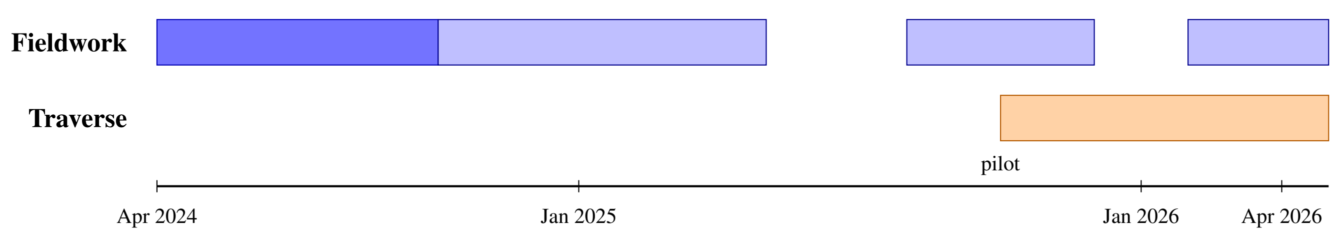

Figure 1: Timeline of fieldwork and Traverse implementation, April 2024–April 2026

3.4 Data analysis

My analysis was inductive and process-oriented, but the aim was to identify recurring mechanisms rather than a universal set of stages. I moved iteratively among fieldnotes, interviews, analytic memos, emerging concepts, causal loop diagrams, and relevant theory to understand how notes were produced, evaluated, used, and repaired across audiences over time. The goal was not to estimate the prevalence of particular attitudes or practices or claim representativeness. Following Merton's view that theory is more than an empirical generalization, I treated the field evidence as the basis for specifying mechanisms and feedback processes, not simply summarizing observed regularities (Merton, 1968). Consistent with process theorizing, I focused on sequences, turning points, delays, and changing relations among actors, artifacts, and practices over time (Langley, 1999).

The analysis proceeded through three overlapping moves. First I examined how clinicians, administrators, and billing professionals produced, evaluated, and acted through notes. Early memos compared what these groups considered a "good note," the purposes they expected notes to serve, and the standards by which notes were judged. The central analytic unit became an episode in which someone judged a note as adequate or inadequate and responded (or didn't) to it.

Second, I compared these episodes across types of use and audience: who detected the problem, how quickly and specifically it was communicated, who compensated, and whether the response changed the note or the work. Third, I traced how clinicians responded when notes did not provide what they needed. I examined what they opened, skipped, searched for, and trusted; when they contacted another person; and when they relied on memory. I then considered how deficiencies were compensated for. Comparing inpatient and outpatient settings provided analytic leverage: deficiencies were more likely to generate visible interpersonal compensation where clinicians used someone else's notes, while clinicians absorbed more of the problem with their own memory when continuity was intrapersonal.

At each stage, causal loop diagrams were used to clarify a causal hypothesis and represent how these relationships fit together. The diagrams were analytic devices rather than quantitative models. They required me to specify the direction and timing of relationships among evaluative feedback, usefulness in subsequent work, reading, authorship, workarounds, and documentation burden. I revised or removed links that were not supported by field evidence (Azoulay and Repenning, 2010; Repenning and Sterman, 2002).

The unexpected introduction of the AI scribe served as a diagnostic perturbation. I was able to compare how clinicians composed and revised notes before and after its introduction across settings. The abrupt change in behavior following its rollout helped clarify the significance of the patterns documented during the preceding fieldwork. It is not evidence of AI's settled effects, nor does it imply that AI caused the patterns that preceded its introduction. The study directly addresses changes in clinicians' production and use of notes; downstream clinical outcomes and longer-run consequences are implications of the model, not completed outcomes observed in the field.

4 Findings

4.1 Searching for the good note: carrying clinical work forward

The most immediate finding was that the note was still important clinically. Clinicians were constantly searching for what they called, in practice if not always explicitly, a "good note." They did not treat all notes as equally useful. They looked for particular notes, written by particular authors, on particular dates, that could make a patient's course intelligible. Amidst consistent complaints about the volume of what more than one informant called "garbage" in the record, in one conversation, a nurse responded to an attending's frustration about the data in the EMR by directing him to "a really nice note" from an outpatient doctor on a specific date. In another case, one pulmonologist said the clinician from the previous encounter didn't mention something he was looking for in the note. A colleague corrected him: "...it does. I couldn't find it either. It's the very last sentence. It's just a tiny mention." These moments were common. Clinicians were not purposelessly opening the record; they were navigating within it to find islands of clinical meaning.

Clinicians assisted each other with search because the record was dense with information but uneven in clinical signal. One provider described the problem as a matter of search cost: "The succinctness is important. If I have to scroll through 20 pages of a past emergency visit to find out like two little bits of information that's not very useful." A primary care provider echoed a similar sentiment, saying:

> It's much harder to find information in a longer note. If you don't have the assessment and plan at the top, it drives me bananas. What it is visually actually really matters. You want it to be quick. You want bullet points. Paragraphs are so noisy and hard to find the information you need.

Both complaints were about the difficulty of finding the clinical signal inside the noise of an increasingly dense record, not about the scarcity of information.

Clinicians struggled to define a good note in the abstract, but they recognized one when they saw it. Across observations and interviews, good notes were described less by formal features such as length, completeness, or template adherence than by what they allowed the reader to recover. A good note conveyed uncertainty, preserved context, and explained why the prior clinician acted as they did. A few clinicians described what they look for specifically:

> If there's a little narrative of medical decision making where it says why they were here, what we did and how they left, that's a lot more useful to me than 20 pages of click boxes where you're kind of trying to search through junk.

This provider hinted at what another said more explicitly, that the important part to them is the narrative rather than the labs, vitals, check boxes, and tables often included as part of a template.

> The less precise the information in the note the better, I mean just personally. I can find the vitals and labs, just give me a narrative, tell me what's going on.

The same primary care provider who complained about the difficulty of finding information in a long note explained that even within her own notes there is a difference between the "good stuff" and the rest:

> The good stuff is always at the top and that's for others and everything that's for me is further down and I know how it's structured and where to find it.

The useful note, then, was not just the complete one. It was the note that made clinical judgment portable. It allowed one clinician's reasoning to be recovered by another, often later and under pressure.

Because good notes were unevenly distributed, clinicians helped one another find them. As above, they conveyed the author or date of a useful note or section of a note to reduce the next clinician's effort in reconstructing a patient's treatment course. When no such note could be found, encounters felt episodic and context-free, often triggering rework. One provider explained how he maintains high-quality notes by saying "many people just copy the progress note into the discharge summary and go…I create mine from scratch to make sure that mine is good." Another expressed a preference for "a new patient with no history" because he could ask the questions himself rather than search through the record. The search itself revealed the important empirical point: the note still sometimes carried the clinical information clinicians needed, but that value was episodic, unevenly distributed, and attached to particular moments of authorship. It was not reliably embedded in the record as a system.

What made these notes good was the authorial work behind them. A note was useful downstream because some prior clinician had taken the time to select, synthesize, and structure the patient's case. In this sense the note was, in Berg's terms, constitutive of clinical work rather than just representational: writing helped organize clinical attention and judgment, and also preserved decisions already made (Berg, 1996; Berg and Bowker, 1997). Clinicians who trained before the widespread use of an EMR described note writing as a moment of synthesis. One older physician recalled dictating charts to a tape recorder and reviewing them the next day:

> Well, you know, it's 8 o'clock at night, I'm home…sitting at my desk, I'm dictating a stack of charts and…you suddenly…you suddenly go 'CLICK! Oh! I got it!' you know, or something you forgot? And then you make a note and you can put it …in your dictation. And when the patient comes back, or if it was something significant, you would call him the next day or have your office call him and say, you know, we think we should do this, or we need to check on that.

The value of writing, on this account, lay in producing an occasion for clinical recognition, not only in preserving a record of care already given.

That synthesis was necessary because the patient encounter rarely presented itself as a clean set of facts. Patient stories were often inconsistent, partial, or contradictory. Providers asked the same question multiple times, from multiple angles, and received different answers from patients, caregivers, or family members. One vascular surgeon described the interpretive work involved:

> And then you'd sit there and try to summarize it in your own mind in a way that made sense. And I didn't always document, you know, I asked this question six times... you try to summarize it in your own mind, okay, here's what I asked. I'm interpreting this to mean this.

Rather than document every contradictory answer, he summarized what he understood the answers to mean. Similarly, after seeing a patient who described a precipitous decline in lung health and recalled running consistently just a few years earlier, a pulmonologist explained to me that he had cared for her for more than twenty years, and that during that time she had not been able to walk up a flight of stairs without her oxygen tank. Her encounter note, therefore, could not reproduce her account of the goings-on and required that he interpret her account differently. A similar dynamic appeared when a neurologist examined a patient who, when asked to lift her right arm, grimaced and made the face and actions of one trying very hard to do so without accomplishing it. Moments later, when asked to move her leg, she used the same arm, without hesitation, to move the bed sheet off of her leg. In such moments, the note's clinical value came not from transcription but from synthesis: transforming observed behavior, patient history, patient narrative, and professional judgment into a usable account.

The search for the good note revealed one test by which clinicians judged the record: whether it carried a prior clinician's selection and synthesis forward in a form another clinician could use. A note could contain the facts of an encounter without meeting this test. The author used the note to render the patient legible; the reader used the note to recover that legibility later. This connection between authorship and use is what made the note clinically meaningful. Yet usefulness in subsequent work was not the only standard applied to the note. The same document also had to stand as an account of prior work before readers who approached it for other purposes.

4.2 When "good" meant different things: two tests of the same note

Clinical notes have long served purposes beyond direct patient care (Kuhn et al., 2015; O'Donnell et al., 2020; Rule et al., 2021; Epstein et al., 2021). At Yankee Health, however, this multiplicity was not simply a matter of having more audiences. It meant that the same note was evaluated according to different standards. Clinicians read notes to understand what had happened and why; coders, revenue cycle professionals who translate documented diagnoses and services into standardized billing codes, read notes to assess whether the record supported a billable claim; clinical documentation integrity staff assessed diagnostic specificity and queried clinicians when documentation was missing, incomplete, or conflicting; compliance officers looked for evidence of regulatory adherence; quality abstractors treated the note as a data source; risk managers and legal audiences read it as a record of defensibility; hospital administrators aggregated it as revenue documentation; and, increasingly, patients, and family members tried to understand and evaluate care.

These audiences wanted different things from the same record, not just *more* documentation. A billing professional later described her group's work as reviewing the record for "missing, incomplete, not clinically supported, conflicting, or unspecified documentation" that could be improved to tell the patient's story more accurately. She emphasized that this work was not only about payment: "we do this first and foremost to tell the accurate story for the patient, because your medical record goes with you everywhere you go." She gave the example of being in a car accident, unresponsive, and dependent on the emergency room physicians having an accurate medical record. "If it's not correct, it's a big problem. It's why we do it." In this account, a good note was specific, complete, clinically supported, non-conflicting, and portable across settings.

The distinction was not between people who cared about patient care and people who did not. The billing professional justified specificity and completeness partly in terms of patient safety, while clinicians routinely adjusted their notes to secure coverage, protect patients, and make their care defensible. What differed was the test being applied to the record. Readers evaluating the note as an account asked whether prior work was sufficiently specified, supported, consistent, and appropriately represented. Readers using it to continue clinical work asked whether it made the case intelligible and carried forward the judgment needed to act.

Clinicians also wanted the note to tell the patient's story, but they often meant something different. In the prior section, clinicians repeatedly described wanting a concise narrative of decision making; they could find labs, vitals, and other structured data elsewhere in the record. A billing professional expressed nearly the opposite:

> "I don't want to have to go hunting for the labs. It's easy with a dot phrase, just put in the note exactly what's happening."

Both positions were reasonable. For the clinician, repeating data already available elsewhere increased search cost. For the billing reader, having that same information in the note made the record easier to review, support, and defend.

Clinicians therefore encountered the note as a daily negotiation among competing definitions of "good," as opposed to some single shared standard. When asked about the purpose of the EMR or of clinical documentation, they described it as being for themselves, for other clinicians, for billing and insurance purposes, for patients and caregivers, or, most often, for some combination of these. One doctor said:

> "People say notes are either for yourself or for others and people have different preferences on how they write it. But for me, it's both."

They didn't argue that any one audience was illegitimate. The problem was that these audiences defined note quality differently. Terminology made this especially vivid. One hospitalist lamented:

> "when I'm just trying to get through my documentation because it's purely a glorified cash register kind of situation, and it's not affecting my patient's care, it's very frustrating to say, okay, is this altered mental status or is this acute metabolic encephalopathy? I mean, we're splitting hairs here. Purely for insurance purposes."

Explaining that the requested terminology means the same thing but isn't a clinically meaningful difference. The providers also try to use these split hairs to their patients' benefit. One explained while documenting post visit:

> "Actually, I'm going to add the symptom of hot flashes. Hot flashes and vasomotor symptoms are technically the same thing, but if they have different ICD codes… I'm just going to try to help her cause with insurance."

These moments certainly didn't show clinicians who were indifferent to billing or who felt billing had no clinical relevance. They showed that the words that made sense for one audience did not necessarily carry the same meaning for another. Clinical documentation was not the execution of a single shared standard. It was the ongoing translation of clinical work into several partially competing languages.

Completeness was the most common site of conflict. For billing, coding, compliance, and legal audiences, completeness often meant that the note contained sufficient evidence, specificity, and traceability to support downstream review. For clinicians, however, completeness could be the enemy of usefulness. A clinically useful note was not necessarily the one that contained the most information; it was the one that made the right information findable. As one provider put it above, "Succinct is key." Even if the note technically contained the relevant data, the clinical question was whether the note organized those data in a way that supported search, reasoning, and action under time pressure. An ICU attending articulated this tension while deciding how to structure his note.

> "I would like to include the blood gas and labs here to make it easier to find, but I guess it's fine… It's in the note somewhere, so it's documented and not a problem for billing, even if it is harder for us to find."

His note satisfied one version of completeness, even if it made the information harder for clinicians to locate and thus use in downstream care. A note could be complete in the sense that the required material appeared somewhere in the document, while still being clinically inefficient because the material was not organized around how clinicians searched, reasoned, and acted under time pressure.

The legal audience sharpened a related problem: a note needed to both contain information, and avoid contradiction. Several clinicians described how dot phrases and linked results made it easy to write "results pending," refresh the note, and unknowingly sign a note with the results in and the text reflecting that they were still pending. The record then became internally inconsistent: formally rich but substantively wrong. After describing this kind of example, one physician called it a "lawyer's dream." Physicians did not dismiss legal defensibility. The tools designed to make documentation more complete could also make it easier to produce a record that looked adequate while containing contradictions a clinician-author had not intended. The same provider went on to say:

> "I used to say the more the better in your note. Putting it in your note should mean that you looked at it, but now with templates and dot phrases and everything, it doesn't. I don't blame the provider, I feel bad for the provider because maybe it was pending when she wrote the note and she refreshed the links and didn't see the results got pulled in."

Patient access introduced a similar tension. Notes increasingly had to be legible and acceptable not only to clinicians and administrative readers, but also to patients and their families and caretakers. This was not inherently problematic; patient dignity, transparency, and comprehension are legitimate values. But patient-facing language could conflict with the shorthand, candor, or diagnostic terminology clinicians used with one another. Multiple providers noted that terms such as "obese," "morbidly obese," or "depression, moderate with delusion," could elicit in-basket messages demanding explanation or revision. One neurologist described pressure to move away from use of the term "psychogenic" in his notes because patients disliked it or felt he was calling them "psycho." One primary care provider put the broader change simply:

> "We have an added layer right now of the patient really reading the note, and I've really had to change how I write my notes. There are things that are not pejorative, but are descriptive, and patients freak out thinking that it is."

This translation work being performed by the providers is not inherently bad. The above provider speaking about pejorative language went on to say "There's positives to it. It has changed the fact that I say the patient declined rather than the patient refused certain things, which is certainly much better." As patients gained more immediate access to their records, clinicians described changing language in anticipation of patient reactions. The same record had to be clinically precise, administratively usable, legally defensible, and interpersonally acceptable.

This section establishes a condition, not yet a decline. The same note was judged along two analytically distinct dimensions. As an account of prior work, it could satisfy or fall short of the standards applied by evaluative audiences. As a resource for ongoing work, it could be more or less useful in carrying clinical judgment forward. The ICU attending's note illustrated one off-diagonal possibility: the needed material was present "somewhere," making the note sufficient for billing while leaving clinicians to search for it. Terminology disputes illustrated the other: language that communicated adequately among clinicians could still invite correction from a payer or objection from a patient. Table 2 summarizes the four combinations these dimensions make possible.

Table 2: Two tests of record adequacy

	Low accountability adequacy	**High accountability adequacy**
Low practice adequacy	Neither useful nor defensible	Formally complete and evaluatively legible, but poor as an instrument of judgment or coordination
High practice adequacy	Useful to workers but unbillable, unauditable, insufficiently documented, or vulnerable to challenge	Adequate for both carrying work forward and holding work to account

Clinicians, as authors of the shared record, absorbed much of the translation work required to satisfy both tests, often through additional documentation time, cognitive effort, and what informants and prior research called "pajama time." The existence of two standards did not itself determine how the note would change or which use would come to govern its form. That depended on what happened when a note fell short under each test. The following sections trace those feedback processes.

4.3 When evaluative shortfalls changed the record

Shortfalls under evaluative standards first appeared to clinicians as interruptions. One hospitalist described a common situation:

> "I'll see the patient and maybe I'll get them on day three of the hospitalization. And by the time I have them, I'm just treating them for something simple like a urinary tract infection. They did not necessarily meet criteria for another diagnosis, but maybe the ED physician said that the person has a history of heart failure. And then all of a sudden, I'm getting this message that says, can you document whether or not this patient has heart failure exacerbation and it's completely irrelevant, but I have to stop everything, review it, document on it... And all of this is purely to determine whether or not we can bill at a higher rate. For the patient... it has nothing to do with the patient's clinical course whatsoever."

The note was judged insufficient under a billing standard, the deficiency was routed back to the clinician-author, and the required response was made actionable through a request for clarification.

Other clinicians described similar translation work. One primary care provider explained as he was writing that a patient's "Hypertension, diabetes, and sleep apnea - well controlled," that: "coding hates this, they want each [condition] listed out. And for each one, one by one to say, hypertension, well controlled, diabetes, well controlled, et cetera, et cetera. I am not redundant, nor do I have the time to be." He said billing would suck it up. Another hospitalist who above characterized the EMR as a "glorified cash register," complained about being asked to split hairs over terminology that meant the same thing clinically, but for insurance purposes mattered. In an ICU, an attending looked at the admission diagnosis:

> "Acute hypoxic respiratory failure... this is a code, not a diagnosis. It's in fact a bucket of diagnoses that give you low oxygen. And when someone is admitted with this, it tells you nothing at all."

The prior clinician had used language that satisfied one evaluative standard but did not carry forward what the next clinician needed to know in order to do the work and treat the patient.

Clinicians knew for whom they wrote, and knew that writing for different audiences required different words, and sometimes they resisted. But this resistance had a price. The same primary care physician who grouped several stable chronic conditions together rather than documenting each in the format billing preferred described a similar tension around an HIV test required by the state in which he practiced. He explained that he would "take the hit" and be downgraded by insurance - paid less and classified as a "worse" doctor - rather than order a test he viewed as clinically inappropriate in that context. The examples clarified the structure in which documentation occurred: professional judgment could prevail locally, but the documented account could still be judged insufficient. Clinical discretion remained possible, but it was no longer free.

The same correction process was embedded in documentation architecture. Providers often used copy-forward to create a new note from a prior note. Copy-forward could support and undermine both dimensions of the record. It reduced rework and helped clinicians preserve longitudinal context across days, but stale information could persist, copied authorship could obscure what happened that day and who performed it, and irrelevant text could accumulate as bloat. In one case, an emergency physician reviewing a discharge summary that still said "MRI Pending" noted wryly that the "MRI was presumably no longer pending at discharge." The text remained in the note, but the account was no longer current or useful, and, in this case, it was wrong.

The concern about the note as a current, attributable, and defensible account was not hypothetical. In a billing meeting I observed before the restriction was implemented, Yankee Health representatives presented inappropriate copy-forward use as a documentation risk. They displayed CMS guidance on copied documentation and explained that copy-forward practices could make it difficult to prove medical necessity or attribute care, contributing to denials. The warning was explicit: if clinicians continued to use copy forward in ways that undermined attribution and medical-necessity support, the organization would have to restrict the function. A few weeks later, this hospital removed copy-forward for hospitalists.

Clinicians had anticipated the problem. One physician said that if copy-forward were removed, "I would riot," explaining that he needed "a running list of what has happened" and that he could not retype the entire hospitalization every day. "People are going to do what they're going to do. Like any tool, it can be used well or dangerously and it's up to the user to make that choice" he said. "You can build barriers, and it's just going to take them more time and make them more miserable to do the same thing." The defense was reasonable: copy-forward allowed clinicians to preserve longitudinal context under severe time pressure. But it also contributed to the very bloat that made notes harder to read and care more difficult to carry forward.

Another clinician made the same point as a theory of why correction directed at account adequacy would burden conscientious clinicians without necessarily improving the record for other uses.

> "People who are not good at writing notes will continue to be bad at this. There's no good stick for this issue of documentation. No one gets hurt. They keep trying to nudge you to write better notes, and the people who care like me change everything and the people who don't care never change. It's a Herculean effort to get people to alter their behavior. It does contribute to burnout because you constantly change your workflow and change how you write and process information and you never get rewarded. And the people who are bad never change and never get punished and even worse get rewarded for other things that they do. I take shortcuts where I can and where I know it won't hurt anything and just do the best that I can. Standardized notes are just hard for the people who care and try and do absolutely nothing for those who don't. Now I'm doing something they don't like... I shouldn't copy things forward, but it saves me 15 to 20 seconds here and there, so I do."

These objections did not prevent the change. Yankee Health had identified a shortfall under an evaluative standard: copy-forward could leave stale text in the note, obscure who had performed the documented work, and weaken support for medical necessity. The system addressed that shortfall by restricting functionality. Yet the clinician's critique proved prescient. To comply, one hospitalist opened an old note on one side of the screen, a new note on the other, and dictated the former into the latter. The workaround introduced rounded values, skipped details, mixed-up laboratory results, and omitted contextual material that did not seem worth reproducing during dictation. A policy intended to make notes more current, attributable, and defensible therefore increased the burden of note production and, in observed practice, could thin the resulting record and introduce new inaccuracies. As another hospitalist put it, the change made it take longer to do the same thing worse.

The case of copy-forward shows why the two dimensions of record adequacy should not be read as quality versus its absence. Yankee Health was pursuing quality by trying to improve the note as an account of prior work: making it current, attributable, and defensible. Clinicians were also pursuing quality by trying to preserve the note as a resource for ongoing work: maintaining continuity, reducing rework, and keeping prior reasoning recoverable. These aims were not inherently opposed. Copy-forward could preserve longitudinal context while also propagating stale or unattributable text; restricting it was intended to improve currency and attribution but made continuity more burdensome and error-prone to sustain.

The consequential difference, then, was not that one use represented quality and the other did not. It was that the intervention was organized around one test of the record. A shortfall identified in the note as an account generated meetings, warnings, permission changes, and system restrictions that altered how future notes were produced. Whether the resulting notes remained useful in ongoing work was a separate question, whose consequences would become visible later and through a different pathway.

A revenue cycle representative who was also a practicing clinician made this logic explicit. His goal, he explained, was to make:

> “sure I have the right words that will give you the right credit for the work that you’re doing. And I tell my physicians when I educate them, you’re kind of in a game, whether you realize it or not, and there’s rules to that game, and it changes, and you don’t have to be privy of it. But remember, there’s CDI teams and coding teams that sort of help you a little bit as you kind of go along.”

Clinicians did not need to understand, or even know, every rule because CDI and coding teams translated those rules into specific requests. He cast CDI not as clinicians’ adversary but as their safety net: it helped physicians receive credit for their work while avoiding exposure in a classification system whose categories they did not control. Yet the safety net was also a correction infrastructure. Care had to be rendered in language that the revenue, coding, and audit apparatus could recognize; when the documented account fell short, the mismatch returned to the author for repair.

Patient access to the record generated a parallel return path, although patients applied different evaluative standards. As patients gained and used more immediate access to their records, clinicians described changing language in anticipation of patient reactions. Patient feedback was fast, emotionally salient, and often tied to messages, complaints, and patient satisfaction consequences, that clinicians understood could affect compensation. One pulmonologist joked, with visible strain, that a patient’s "BMI is high so I can click obese without fear of retribution," referencing the increase in messages that could follow language a patient disliked. Clinicians again performed the translation work, trying to preserve clinical precision and usefulness while anticipating the standards of intelligibility, respect, and fidelity that patients and caregivers would apply to the account.

Across these examples, a shared feedback property united evaluative audiences: judgments of insufficiency could travel back to an identifiable author. Billing and CDI shortfalls returned as queries to individual clinicians; when repeated queries became visible as a pattern, they prompted training, warnings, and restrictions on system functionality. Patient-facing concerns returned as messages, complaints, and patient-satisfaction consequences. These standards were not illegitimate. Notes should appropriately support payment, demonstrate medical necessity, avoid stale copied text, and remain understandable and respectful to patients. Nor were the standards equivalent. What their feedback processes shared was that perceived shortfalls were visible, attributable, and actionable.

Failures discovered when clinicians tried to carry work forward had no comparably reliable route back to the author. Clinicians could complain that a note was too long, redundant, or no longer useful for care, but such judgments rarely produced the same specific and immediate response as a query, denial, or patient complaint. More often, the next clinician absorbed the deficiency through additional search, verification, or communication. The asymmetry therefore lay in whether their judgments could reliably change the record and its future production. This section therefore establishes the first feedback process: asymmetric correction, evaluative shortfalls altered the record, while failures in use were more often absorbed by the work that followed.

In Figure 2, I begin to formalize these findings with a causal loop diagram. The model treats two properties of the record as central stocks or levels that evolve through accumulated feedback over time. *Account adequacy* captures how well the record satisfies the standards applied to it as an account of prior work; *clinical usefulness* captures how well the record supports ongoing clinical work. For economy, *account adequacy* summarizes adequacy under prevailing evaluative standards. It does not imply consensus among evaluative audiences: billing, legal, administrative, and patient audiences applied different standards. What they shared was that judgments of insufficiency could be returned to an identifiable author and made actionable.

Asymmetric correction consists of two formally parallel but operationally unequal balancing feedback processes acting on these stocks. The first, **Evaluative Correction, B1**, functions largely as intended. When *account adequacy* falls below *desired account adequacy*, the *account adequacy gap* widens. That gap increases *evaluative correction pressure*, which produces *evaluative correction* through training, interface changes, queries, required fields, data validation, or workflow restrictions, bringing account adequacy back towards the desired level.

4.4 When failures in use changed the work

When an evaluative audience judged the note insufficient, the deficiency tended to change the record. When the note failed as a resource for ongoing work, clinicians more often changed how they worked. Clinicians still opened the chart, searched the chart, and acted through the chart. What changed was their expectation that the note itself would contain the clinical reasoning they needed. This shift preceded the AI scribe. Before clinicians could ratify AI-generated notes, they had already begun to treat clinician-authored notes as unreliable carriers of clinical meaning. The change did not appear as nonuse. It appeared as skeptical, selective use: clinicians read around the note.

Clinicians described this stance as verification rather than trust. One physician explained, “I trust my own note. So that’s… that’s where the source of truth [is] for me. I don’t trust anyone’s notes.

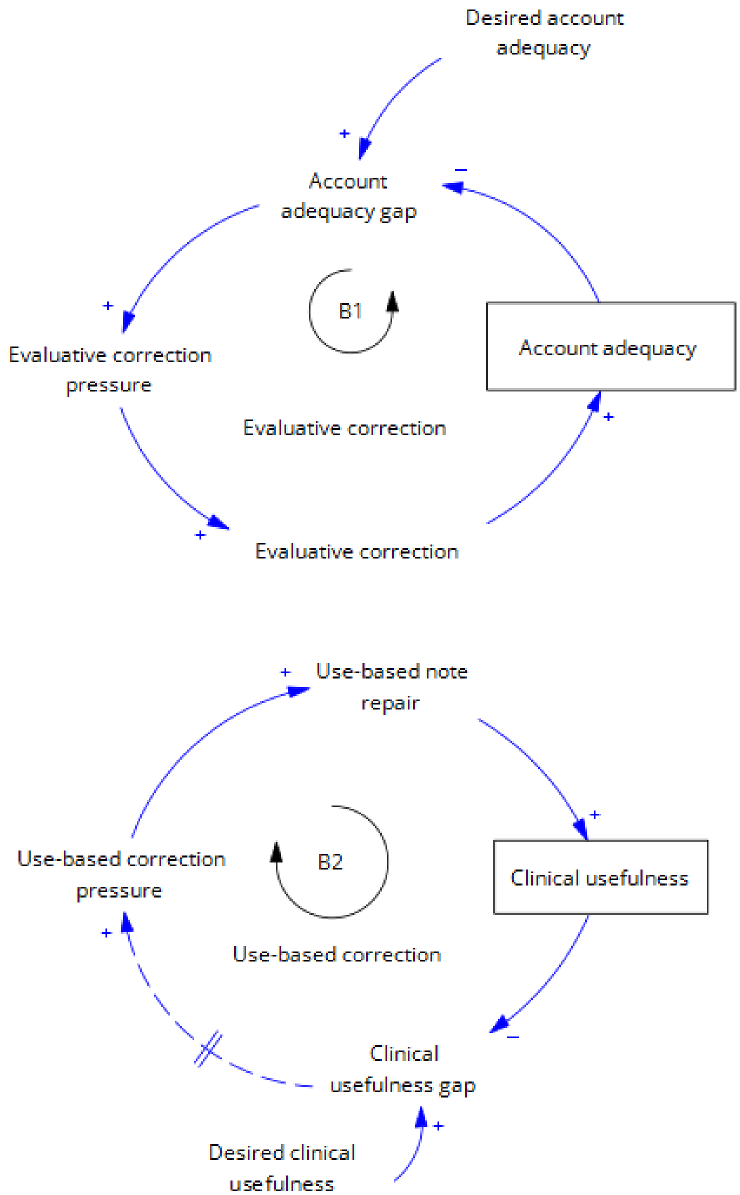

Figure 2: Central stocks and balancing loops. *Note*: Arrows indicate the direction of causality. Signs ("+" or "-") at arrowheads indicate the polarity of relationships: a "+" denotes that an increase in the independent variable causes the dependent variable to increase, while a decrease causes a decrease. Similarly, a "-" indicates that an increase in the independent variable causes the dependent variable to decrease, while a decrease causes an increase. The loop identifier, B, indicates a balancing feedback loop. Balancing loops are goal seeking and thus self-correcting (see (Sterman, 2000)).

That's sort of how I think about it. But I will verify everything they put on... on that record." Even that trust in his own note was partial. He went on to describe finding mistakes in others' documentation and sometimes in his own:

> "It turns out that that was a mistake, by the way, by the radiologists. So that happens, and I talked to the radiologist afterward. Obviously it is... very rare that that happened. Like, probably 1 in a million, I'm thinking, or, you know, hundred thousands. But I do find that mistakes happen, and even on my own note, even though it's the gold standard, I find myself, when I'm looking at the previous notes, sometimes they are a mistake that I made."

Another clinician contrasted the electronic record with handwritten notes, explaining that technology added a layer of skepticism:

> "Because there are so many errors in the chart, I can't take the note as gospel. There's two options. One, the doctor said one thing and the patient didn't hear it or didn't understand it or thought they said something else. Or two, the documentation is wrong. And so either of those options means I just have to do more digging. And it becomes a bit of a digging expedition. If the note is handwritten, they had to write it, so it's truth. But the technology necessarily adds a layer of skepticism because you can always have a typo or autocorrect or accidentally copy forward something that doesn't belong to that patient. A data point is a data point. A lab's a lab, but you can't even believe a lab sometimes."

Clinicians changed how they used the record. Rather than treating the note as a self-contained guide to what had happened and why, they reconstructed the case from the note and other traces in the chart. No longer beginning with the most recent note and reading backwards, clinicians often moved directly to sections they regarded as more reliable or efficient: pathology, imaging, laboratory trends, medication orders, impressions, or the assessment and plan. An oncologist, frustrated by the volume of non-oncology information, explained:

> "that become[s] very hard, and I have to sort of dig through the chart to understand the reason why. In those cases, I go straight to pathology."

Pathology couldn't substitute for clinical reasoning, but it could provide a more reliable starting point for reconstructing the case. On another day, a specialist pulled up a primary care note, called it "useless," and scrolled for several minutes through check boxes for Medicare, vital signs, and other auto-populated material before giving up. "I can't find the section where he just tells me what's up with this patient? We can't get anything useful from this." After giving up he said he would have to "go back to the med school way of things" and get the history from the patient himself. On another occasion the hospital's image viewing software was down and an APRN commented to me "I'm having to read these notes in detail because I can't see the imaging." The note had become a fallback rather than the default location of useful clinical synthesis.

At first, I interpreted this pattern as a shift in preference from qualitative narrative to quantitative or discrete data. Closer analysis showed otherwise. In several observations, physicians searched laboratory orders for evidence of what another clinician might have been thinking. One physician explained that ordering a particular, potentially nonstandard test could reveal what another doctor had considered. Clinicians used discrete data as traces of absent reasoning. They did not abandon the need for narrative clinical judgment; they displaced the search for it into a more uncertain forensic exercise. The record remained information-rich, but the written synthesis that made information clinically usable became harder to recover (Berg, 1996, 1997; Feldman and March, 1981).

The absence of clinical feedback on the note itself was striking, especially given how routinely they noticed problems in the notes. When readers applied evaluative standards, shortfalls generated queries, warnings, and correction requests. When clinicians encountered a deficiency while trying to continue clinical work, they more often compensated for it themselves. The asymmetry lay not in detection but in return: shortfalls identified through evaluation changed the record, whereas failures encountered in use more often changed the work. One physician explained to me that:

> “You know, I don’t hear from other people a heck of a lot. When I’m asked to see somebody or they send me a communication is... it’s Monday, I last saw the patient on Friday. It wasn’t on call over the weekend. It’s a "Hey we’re wondering if Mrs. Smith can go. Can she be discharged?"... I don’t usually get a lot of someone calling me and saying I looked at your note and I wanted you to explain this to me why you put that in your note or what you’re... I want you to go into the thought process of that with me. In more depth. I don’t usually get those kind of calls."

The communication is about clinical judgment and what to do now, not about improving the record that had failed to carry the reasoning forward. Another clinician made the contrast explicit when asked about the feedback he gets on his notes.

> “That’s what the clinical documentation team is. They want to clarify documentation for purposes of billing and reducing the denial rates for Medicare.... But we don’t get direct feedback on notes from [doctors].”

He gave feedback to residents on what belonged in the note and why, but described this as part of their medical training, rather than an organizational feedback process. Outside of the training relationship, he said,

> “I get no feedback on what I’m doing and how that affects people." Others echoed the same curiosity and absence: "I’m super curious what… what people think of my notes If they’re helpful or not... Everyone says, oh my gosh, I’m so curious, but no one’s ever told me how they feel, no one gives me, uh, any feedback.”

The second balancing loop in the bottom half of Figure 2, **Use-Based Correction, B2**, is formally parallel to B1 but operationally much weaker. When *clinical usefulness* falls below *desired clinical usefulness*, the *clinical usefulness gap* widens. In principle, a wider gap should generate *use-based correction pressure*, prompting *use-based note repair* and thereby increasing clinical usefulness and narrowing the gap. Such repair could take the form of an addendum, direct feedback to the author, or a change in how subsequent notes were written. In practice, however, failures in use rarely generated a timely corrective response. I depict the link between the *clinical usefulness gap* and *use-based correction pressure* as both weak, with a dotted line, and delayed, with double hash marks. Direct feedback about whether a note supported ongoing work was concentrated in training relationships and rare in routine peer work. Moreover, a downstream clinician might not encounter the note for months or years after it was written. Even then, the clinician more often compensated locally for the deficiency than returned it to the author.

B1 and B2 were therefore structurally parallel but operationally unequal. On the evaluative side, standards defining *desired account adequacy* were more often explicit, attached to identifiable audiences, and connected to established response routines. On the use side, *desired clinical usefulness* became apparent in future episodes of work, was distributed across downstream readers, and was rarely converted into timely, attributable feedback. Importantly, the two loops did not correspond to fixed types of textual error. The same stale diagnosis, contradiction, or missing rationale could activate *evaluative correction* if it was judged against an evaluative standard, or *use-based note repair* if it impeded ongoing work and the reader returned the deficiency. More often, however, the reader compensated locally, leaving the clinical usefulness gap open without activating B2.

The weakness of B2 created the conditions for the first reinforcing process, which I call *recursive withdrawal*. When failures in use did not generate note repair, declining clinical usefulness reduced *clinician reliance on the note*. Lower reliance reduced *anticipated clinical readership*; lower anticipated readership reduced *clinician authorship investment*; and lower authorship investment further reduced clinical usefulness. As readers, clinicians expected less from notes and therefore verified, inferred, skipped, or bypassed them. As authors, they expected others to do the same and therefore had less reason to invest scarce time in selecting, synthesizing, and carrying clinical judgment forward. Recursive withdrawal did not require anyone to stop caring about patients. It required only that clinicians adapt their behavior to a record they no longer expected to carry the judgment needed for ongoing work reliably.

One physician described the professional accommodation that made this process possible:

> "I think, in this profession, people kind of understand what each of us is going through, so if I send someone in to get their right rotator cuff evaluated and it's actually their left… It's pretty rare for that person to actually, kind of. You know, get clarification for that. They'll just kind of see what's in front of them and make do with it. There's almost a certain amount of, um… acquiescence."

This is a professional accommodation, a collegial tolerance, in response to widely experienced overload, not indifference to care. But it meant that failures encountered in use rarely traveled back as corrections to the record. They were absorbed in the next clinician's work. On more than one occasion, I saw physicians note mistakes in the record aloud, but when asked, there was no intent to message the provider or addend the record. They would simply endeavor not to make the same mistake in today's note – even when the prior mistake had been their own.

There were exceptions and they clarified some of the missing structure. One provider saw that he had used the wrong name in a patient's prior note. He expressed dismay and went to addend it after telling me it "looked bad." I asked to whom and he explained how bothered he would be if his own physician used the wrong name in his record. Although legal and billing audiences could contest the adequacy of his account on multiple grounds, what was salient was making the patient feel cared for. Another physician described actively sending Odyssey messages to clinicians:

> "If it's relevant… I will just send an Odyssey message to the clinician and say, hey, I saw Mrs. Jones today, and she said you told her to leave this dose the same, but that's not what your note says. So I will absolutely, especially… if they're in our network, it's easy. So I will do that. If we get a radiology report that right-left's mixed up breast cancer, whatever. I'll have someone on the team call the radiology department to get that report amended. You gotta fix these errors, because if you don't, they just get regurgitated until infinity. And so, I… again, the cut and paste stuff, you know? So I absolutely will ask clinicians, ask radiology to fix reports. If it's a clinician outside of our network, it's more challenging to do that. And, and in terms of feedback to the primary care about errors. I don't think anybody does it. I've never… I've never had."

The limits to this repair are explicit. It is much easier when the other clinician is in network, and while she tries to give this feedback herself, she in primary care has never received it, in spite of acknowledging that there were surely mistakes in her notes somewhere. The historical record could be repaired but depended on individual effort, proximity, and relationship. Feedback from downstream use was not institutionalized.

Reader withdrawal then fed back into authorial practice. One physician discussing stale language in ICU notes, remarked,

> "Honestly, no one even looks at the stuff, so who cares if I have sedated and intubated for most people in the ICU, and I forget to delete it and add awake and alert… That's a stupid mistake, but no one cares."

The comment was partly self-critical, but it also revealed a changed expectation about readership. If no one would read that part of the note closely, maintaining it as if it would be read no longer seemed worth the effort. The risk, however, was not imaginary. A few days later, a consultant in the same ICU pre-charted that a patient was intubated and, upon seeing that the patient was not, asked me in dismay "Didn't we see somewhere in the chart that this patient was intubated? Did I make that up?" What one author treated as low-consequence residue became another reader's uncertain evidence.

The same relationship between anticipated use and production effort appeared in smaller acts of data entry. Clinicians used an emic term – "buttonology" – for the practical art of clicking whatever needed to be clicked in the interface so that care could proceed. Rather than an indication of laziness, buttonology reflected a system that required an entry even when no accurate or clinically meaningful response was available. In one observation, a doctor tried to complete a required field asking for the date of a prior pap smear. The patient was seventeen and had never had one. The system would not allow the field to be left blank, marked not applicable, filled with zeros or entered as the current date. The doctor spent a number of minutes endeavoring to get around the system, and after trying to satisfy the system accurately, he entered an arbitrary date so that he could close her record and proceed with her care. He typed into her note that the pap smear date should be ignored as it was system-required. The interface's completion rule had been satisfied, but the resulting entry reduced the record's adequacy both as an account of prior work and as a resource for later use. Satisfying one machine-enforced criterion had created deficiencies under other evaluative and practical standards.

When a colleague walked by his office a few minutes later he asked if she had encountered this issue before. Indicating that she hadn't, he explained what he did and she expressed appreciation for the knowledge of how to work around it. This sort of information was frequently passed clinician to clinician. What traveled was knowledge of how to bypass the requirement, not feedback capable of changing the requirement or repairing the resulting record.

A primary care provider showed the same pattern at a smaller scale. Changing a medication despite a pop-up warning, she clicked "risk over benefit." When I asked why she replied that it was simply faster because it was the first option in the dropdown. Her presumption wasn't only that accurate data didn't matter in principle. The click completed the required workflow without necessarily producing an accurate or interpretable account of the decision. As another clinician put it: "I know it's important and communications are important as someone who drops in and out and accurate data is huge, but this is just so easy." When clinicians did not expect a field, phrase, or section would be meaningfully used, the expected return to careful entry fell, making speed a rational response to burden.

Clinicians recognized the tension between what they could produce under pressure and what they wished the note would be. In one case, after copying forward his own prior note, a physician paused and said that he could leave it as it was because it was "fine," but that it did not look or sound like him, so he added another dictated paragraph. This moment shows that *clinician authorship investment* had not disappeared. Clinicians still sometimes invested in restoring voice, judgment, and synthesis to the note. But doing so had become discretionary, effortful, and vulnerable to abandonment under pressure. On that day, the physician explicitly noted that he had been unusually productive to "look good" for me and therefore had written better notes. The "good note" was still possible, but it increasingly required protected attention rather than being the ordinary output of documentation.

The earlier search for the "good note" thus became evidence of broader withdrawal. Clinicians still wanted notes that made another clinician's thinking recoverable. But the need to search for such notes revealed that *clinical usefulness* had become episodic rather than expected. When no good note could be found, clinicians verified, inferred, reconstructed, and supplemented. These practices allowed care to continue, but they also changed the note's place in the system. The note remained organizationally mandatory, but it was no longer the default location of clinical meaning. When an evaluative audience judged the note insufficient, the record changed. When the note failed in use, clinicians changed how they worked around it. As clinical usefulness became less reliable, clinicians relied less on the note as a guide, even when they spent more time searching the record. Lower reliance reduced their expectation that colleagues would read and use what they wrote, and lower anticipated readership reduced their investment in writing for future clinical use. Each adjustment made the next more reasonable.

Represented graphically in Figure 3, **Recursive Withdrawal, R1**, explains declining *clinical usefulness* through a reinforcing relationship among use, anticipated readership, and authorship. As *clinical usefulness* declines, *clinician reliance on the note* falls. Lower reliance reduces *anticipated clinical readership*: because clinicians themselves no longer treat notes as dependable resources for ongoing work, they have less reason to expect colleagues to do so. Lower anticipated readership reduces *clinician authorship investment*, i.e., the attention devoted to selecting, synthesizing, and writing for future readers. Lower authorship investment, in turn, further reduces clinical usefulness. The loop also operates in the strengthening direction: when notes are clinically useful, clinicians rely on them, anticipate that colleagues will read them, and invest more in making their judgment recoverable. Recursive withdrawal names the declining branch of this self-reinforcing structure. Nothing in this loop requires *account adequacy* to decline: the note may continue to satisfy prevailing evaluative standards while becoming progressively less useful for ongoing clinical work.

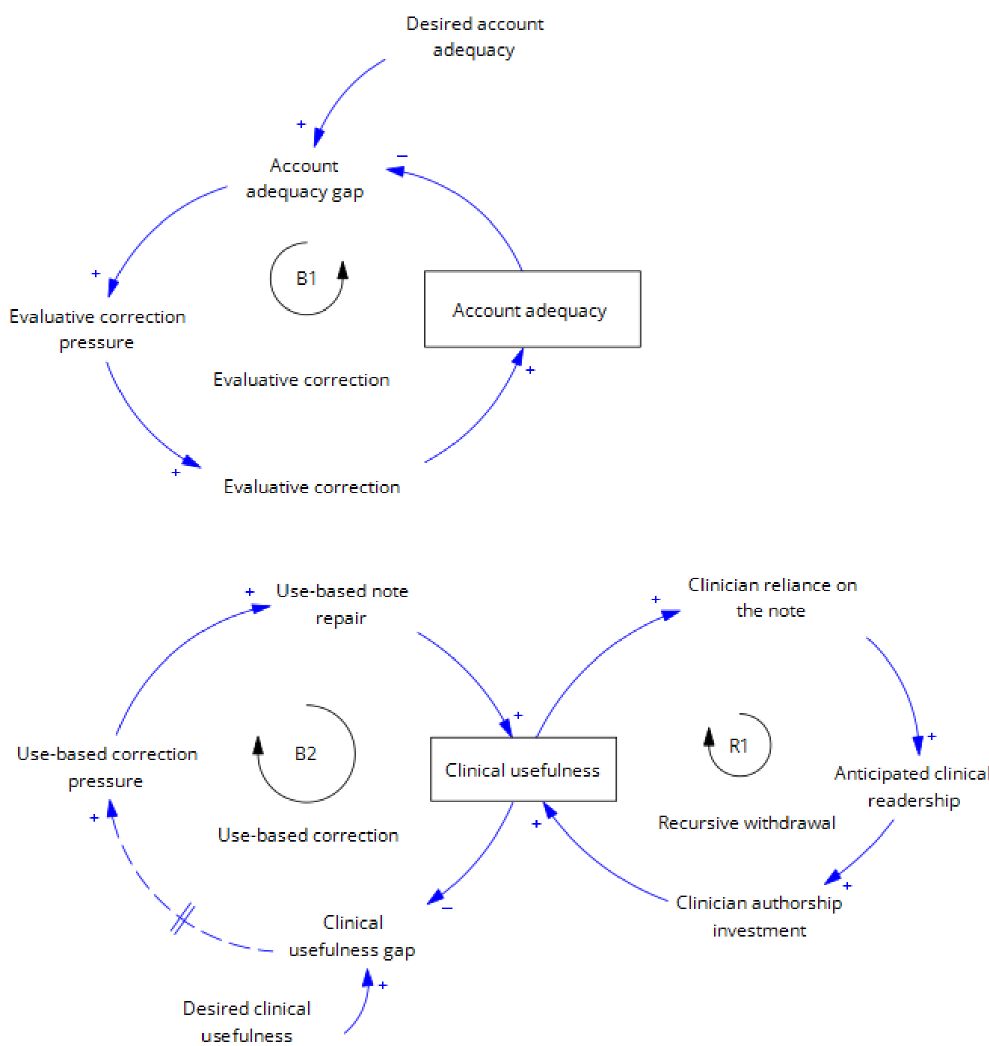

Figure 3: Reinforcing loop 1. *Note*: The loop identifier, R, indicates a self-reinforcing feedback. See (Sterman, 2000).

4.5 Moving coordination elsewhere: preserving local care, weakening shared memory

Recursive withdrawal did not mean coordination stopped. It meant coordination moved. For clinicians, the growing imbalance between the note's adequacy as an account and its usefulness for ongoing work was not abstract; it appeared in the work required both to produce and to use the record. In the model, I define *documentation burden* as the time, attention, and effort required across these activities. Production costs included composing, translating, correcting, and repairing notes; use costs included searching, verifying, and reconstructing what the note did not make readily available. Burden names the cost of this work, not a claim that the work produces no value. I model the two components together because they are recursively intertwined: an intervention that reduces production costs for one actor may increase use costs for another, and clinicians may reduce the burden they experience by bypassing the note while shifting costs to downstream readers.

At Yankee Health, evaluative correction, for example, could increase both components of documentation burden. It required authors to translate, clarify, correct, or restructure the account and could add material or impose forms that made clinically relevant information harder for readers to locate. Declining clinical usefulness further increased the work of searching, verifying, and reconstructing. Clinicians responded by coordinating through other channels and repositories: conversations, texts, TigerTexts, emails, calls, SecureChats, Teams messages, patients, handoffs, formatting workarounds, relationships, and memory. These substitutions preserved local care continuity, but they also deepened and stabilized withdrawal from the note. The more clinicians solved coordination problems elsewhere, the less immediate pressure there was to repair the note as a resource for ongoing work. The note remained mandatory, but its role as shared organizational memory weakened.

The first way clinicians solved coordination problems elsewhere was with direct communication. An oncologist, describing shared care with a radiation oncologist, explained that it:

> "might be important to just look at my note to ... make sure that we're all on the same page. But oftentimes, that... is a conversation or an exchange, a communication outside of the note. The note is really just to document, sort of, my thought..."

He went on to explain that he did not routinely read the radiation oncologist's note to verify that the patient was receiving the correct radiation; he trusted the colleague's role. But he also recognized the importance of his note when a patient went to the emergency department, as there the oncology note might become the "source of truth" for clinicians who don't know the patient. Even in that case, however, he described the note as a starting point. A reader might use it to understand the cancer context but then call him to confirm "what exactly [is] happening."

When coordination occurred within an ongoing relationship, clinicians could rely on conversation, role trust, or direct exchange. When coordination had to travel across people who didn't know one another, the record was supposed to carry that context. But because they had already learned that the note did this poorly, the record's weakness increased the value of knowing whom to call. In one outpatient observation, a pulmonologist reading a radiology note that mentioned cancer paused because he remembered that the patient had never had cancer or radiation. “Why did they mention that? … Maybe I should check with the radiologist. I know the guy." He then messaged the radiologist, whom he knew, to clarify. Upon receipt of the answer that the note was mistaken, he updated his mental model of the patient but did not change the radiology note or addend his own. The error had been corrected for the present encounter, but not the shared record.

Clinicians also used side channels immediately after writing notes. After writing a note for a referred patient, a pulmonologist immediately opened his phone and sent a regular text to the ENT physician to say he was sending a TigerText. He then opened TigerText and requested a discussion about the patient. He told me he didn’t know if the other provider would even read his TigerText which is why he supplemented his note doubly with a regular text. A half hour later, they spoke by phone about context the ENT would need for treatment. The note had been written, but the clinician still did not trust it to carry the coordination work on its own.

These practices showed that clinicians still needed the coordination work the note no longer reliably supported, not that they didn’t care about the record. Research on workarounds shows that frontline workers often develop locally rational adaptations that allow work to continue while making the conditions that necessitated those adaptations harder to see and repair (Tucker and Edmondson, 2003; Morrison, 2015; den Nieuwenboer et al., 2017). At Yankee Health, reliance on workarounds and coordination outside the note preserved local care continuity when clinicians could no longer rely on the note alone to communicate what mattered.

Sometimes the substitute was another clinician; sometimes it was the patient. Clinicians, of course, routinely ask patients to recount their histories. But in my observations, they often did something more specific: they used the patient as a faster alternative to searching the record for information that should, in principle, have been available there. Physicians asked patients when a procedure had occurred, what medication they were taking, whether they had received a prior treatment, or what dose they had been prescribed. If the patient answered with confidence, clinicians often accepted the answer and moved on; if the patient hesitated, they returned to the record to verify. This confidence-contingent verification was a practical response to a dense, low-signal chart, allowing clinicians to avoid another extended search when the patient could provide an adequate answer. Further, medicine is only getting more complex. In one observation, an NP who was looking for information in a note stopped and said, “I guess I can just ask him rather than searching around the chart like an idiot.”

The patient became, in that moment, a local substitute for the record: a more immediate but less durable way to recover information the chart should have made accessible. Like other workarounds, this preserved local care while leaving the underlying deficiency unrepaired. The clinician obtained the answer needed for the encounter, but the record did not necessarily become easier for the next clinician to use. Because the immediate problem had been resolved, the failure in use also generated little visible pressure for note repair.

The episode further illustrates why the distinction developed in this paper concerns uses rather than fixed audiences. Elsewhere, patients evaluated notes as accounts of prior work; here, they became resources clinicians used when the note failed to carry work forward.

Other workarounds stayed inside the record but outside the encounter note. When the note had become too long or too noisy to reliably guide a reader, they tried various ways to indicate what mattered most to carry work forward: bolded sentences, paragraphs at the top with the most important information, altered fonts or colors, or tables of key results. One provider sent an email with the subject line "I KNOW I WROTE IT IN THE NOTE BUT..." followed by the information she worried a consultant would miss. The humor of the subject line depended on a shared understanding: writing something in the note was no longer enough to make it communicatively effective. The note remained the formal location of documentation, but clinicians increasingly used additional signals to make clinically useful information visible.

Handoffs worked similarly. The third and final time that I shadowed a PA, a full year after the first time, I commented that it was the first time I had seen him use the handoff field in the EMR. He explained that yes, he was trying something new that had worked for him personally.

> "I do it because I appreciate it. So, I try to do it for other people. The guy tonight is another per diem person, so I want to help him out, so he knows what to do with the patient when he gets here."

The handoff field, a free text field on the patient record that was not attached to an individual note or accessible by patients, insurance, or any other party, was rarely used and inconsistently read; I only saw it a handful of times in my years in the field. But it was an emerging way that clinicians tried to convey what the note didn't, "what to do with the patient." It helped the next clinician act, but it did so by relocating coordination away from the note itself.

The form of substitution varied with the continuity structure of the work. Where continuity crossed people – across shifts, services, or settings – withdrawal from the note became ever more visible as *reliance on coordination outside the note*: texts, calls, huddles, hallway conversations, and direct messages. One PA in the ICU described the hierarchy clearly when discussing consultant guidance for medication dosing: "For most consultants, we work at their discretion. If they don't tell us via tiger text or stop by and tell us, then we'll scour the record." The shared record was not the expected first channel. It had become the fallback when the preferred form of communication did not arrive.

In Figure 4, the second reinforcing feedback mechanism, **Coordination Elsewhere, R2**, shows how documentation burden can redirect coordination away from the note and thereby further reduce its *clinical usefulness*. The correction work undertaken to increase *account adequacy* enters this process through *documentation burden (use cost + production cost)*. Evaluative correction can increase production costs by requiring clinicians to translate, clarify, restructure, or repair the documented account; it can also increase use costs when the resulting material makes clinically relevant information harder to locate. Declining clinical usefulness separately increases documentation burden by requiring readers to search, verify, and reconstruct what the note does not make readily available.

As documentation burden rises, clinicians increase their *reliance on coordination outside the note* through calls, messages, handoffs, patients, relationships, and memory. When the selection, synthesis, and context generated through those interactions remain outside the note, its clinical usefulness declines further. Lower clinical usefulness then raises the use-cost component of documentation burden, prompting still greater reliance on coordination elsewhere and closing the reinforcing loop. **R2** is therefore locally adaptive but systemically erosive: coordination continues and local care is preserved, but the shared note carries less of the judgment that would make subsequent use easier.

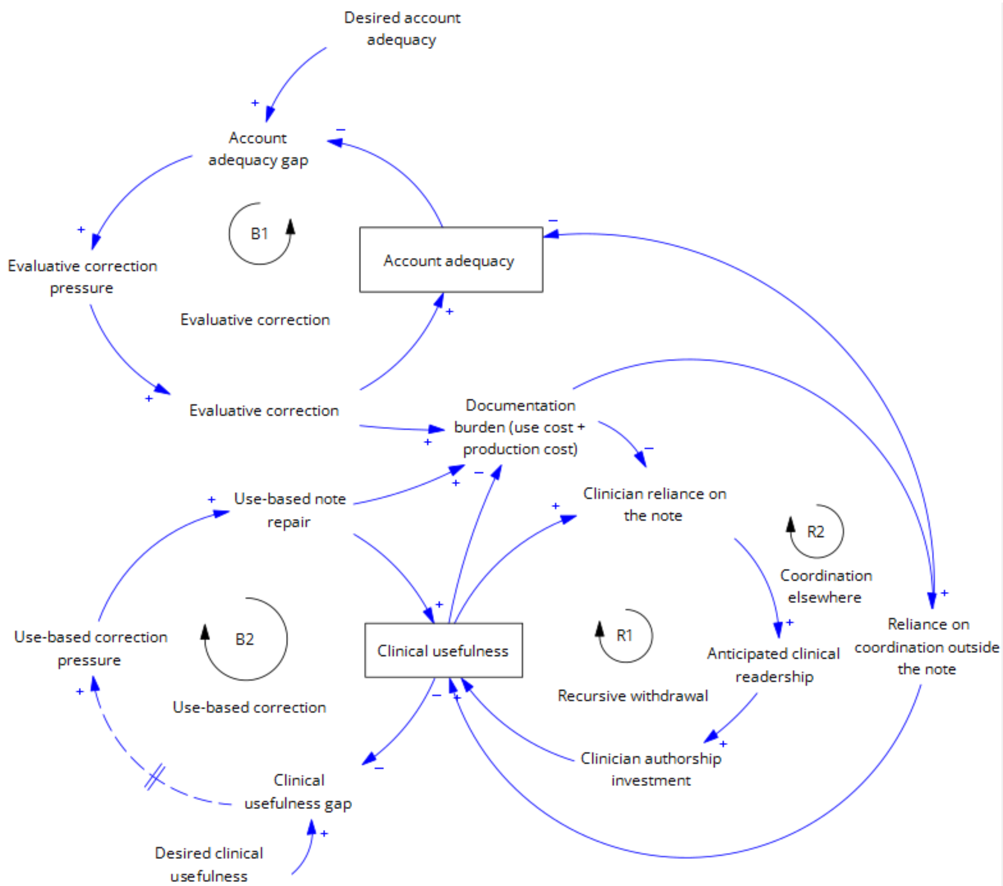

Figure 4: Reinforcing loop 2.

The shadow system became visible to the organization primarily through proxy measures rather than as direct evidence of failures in use. In this case, it appeared as communication volume. During a quarterly hospitalist meeting, the increasing number of daily TigerTexts became an agenda item. The hospitalist leader asked who the “culprits” were and “what time in the morning do they start contacting you.” The message volume was a symptom of coordination displaced from the record into an expanding informal system. A hospitalist remarked after the meeting that another administrator had once asked them to keep track of their TigerText volume. He lamented that the only intervention that had done anything was to get the nurses to stop saying “thanks.” The issue appeared as message overload, but less visible was what the messages resulted from and substituted for: a record that no longer reliably carried context.

Recursive withdrawal was therefore easier to see in some continuity structures than others. In inpatient and hospital-based care, declining *clinical usefulness* generated visible shadow traffic as coordination moved into meetings, calls, messages, and hallway conversations. In outpatient and longitudinal care, the same decline could be mistaken for a healthy record system because little visibly broke.

Where continuity remained largely within one clinician, *reliance on coordination outside the note* often took the form of memory. Primary care physicians and longitudinal specialists could rely on their recollection of the patient, their own interpretation of medication lists and laboratory values, and repeated encounters to scaffold what the note did not provide. A hepatologist seeing a patient he knew well noticed that the charted dosing seemed wrong because he remembered the patient's regimen. A primary care provider noticing osteoporosis was missing from a woman's chart said "Forgive me, I think I remember you have osteoporosis" and added it.

In these cases, memory preserved *local care continuity*, but it did not repair the historical record. In all my time in the field, I observed a prior mistake result in an addendum to the historical note only once. Memory also transferred poorly, scaled poorly, and could not survive the clinician's absence. The most complete version of the patient's story increasingly resided not in the shared record but in one physician's head. The immediate failure in use had been overcome, but little signal of that failure traveled back to the record or the organization.

Workaround substitution therefore stabilized recursive withdrawal. Failures encountered in use were compensated for outside the note. This compensation allowed care to continue, but it also reduced clinician reliance on the note, weakened *use-based correction pressure*, and lowered the expected return to *clinician authorship investment*. Side channels helped clinicians act despite the note. Patients and memory helped clinicians bypass it. Formatting tricks and handoff fields helped them signal within and around the note, but still in the system. Each substitute was locally reasonable.Together they made the note's declining usefulness in ongoing work less visible to the organization and less urgent to repair. This allowed the record to remain mandatory and responsive to prevailing evaluative demands even as coordination increasingly depended on clinical judgment and synthesis the note did not contain.

The model's third reinforcing loop, **Local Compensation, R3**, formalizes how local success can sustain system-level deterioration. Greater *reliance on coordination outside the note* increases *local care continuity* because clinicians compensate for what the note no longer supplies. Successful local compensation, however, reduces the *organizational visibility of failures in use*: the same activity that protects present care conceals the deficiency from actors positioned to change the record. Lower visibility weakens *use-based correction pressure*, reducing *use-based note repair* and allowing clinical usefulness to decline further. As clinical usefulness declines, *documentation burden* rises, prompting still greater reliance on coordination outside the note and closing the reinforcing loop. Nothing in this loop requires *account adequacy* to decline; the record may continue to satisfy prevailing evaluative standards while becoming progressively less useful for ongoing clinical work.

Two further connections capture how the organization's existing measurement and correction infrastructure shaped what it could see and how it could respond. The first is a negative link from *reliance on coordination outside the note* to *account adequacy*. As more clinically consequential coordination occurred through calls, messages, handoffs, patients, relationships, and memory, some of the context, decisions, and attribution needed to account for prior work remained outside the required record. The same practices that preserved local care could therefore make the note less complete or defensible as an account. This link does not imply that every side conversation should be documented.

The second connection runs from *organizational visibility of failures in use* to *desired account ade-*

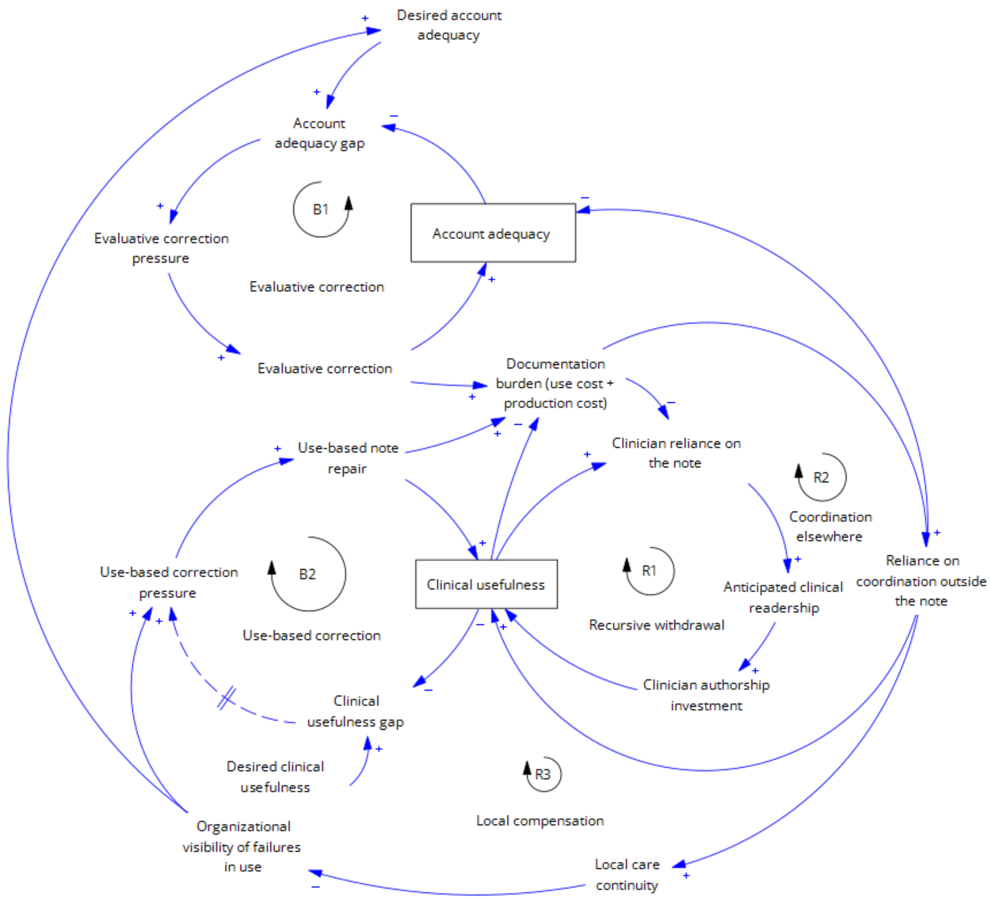

Figure 5: Reinforcing loop 3, Local Compensation.

quacy. When low clinical usefulness became visible, it usually appeared through proxies—message volume, delays, nonstandard communication, or departures from required workflow—rather than as direct evidence that the note had failed to carry clinical judgment forward. The organization therefore tended to respond through the evaluative and design mechanisms it already possessed, raising what it required of the account. This was not because the organization was indifferent to clinical usefulness, but because it had a more developed infrastructure for specifying, attributing, and correcting evaluative shortfalls than for converting failures in use into *use-based note repair*. A signal originating in low clinical usefulness could therefore widen the *account adequacy gap*, increase *evaluative correction pressure*, and add *documentation burden* without directly repairing the note as a resource for ongoing work.

Table 2 described four possible combinations of *account adequacy* and *clinical usefulness*; the feedback processes explain how the record moved among them. *Evaluative correction* tended to increase account adequacy under the standards that activated correction. *Use-based note repair*, by contrast, remained weak and delayed because failures in use were more often absorbed through coordination elsewhere and local compensation than returned to the record. Recursive withdrawal then reduced clinician reliance on the note, anticipated clinical readership, and clinician authorship investment, further weakening clinical usefulness. Together, these processes produced *differential coupling*: the two dimensions did not merely vary independently but became governed by feedback systems that differed in strength, speed, and delay. The resulting condition, a note that remained adequate enough under prevailing evaluative standards while becoming less useful and more burdensome in ongoing work, was not a stable type of record but an outcome repeatedly reproduced through unequal correction.

4.6 Nothing's obviously wrong

During the final phase of my fieldwork, Yankee Health incidentally deployed Traverse, an ambient AI-based note writing tool, or AI scribe that that recorded clinical encounters and generated draft encounter notes within moments. Among the clinicians I observed, its reception was strikingly positive. Most clinicians described being more present with patients, finishing notes faster, leaving work earlier, and avoiding hours of after-hours documentation. One exultant provider said simply "AI has changed my life." Traverse also solved real coordination problems. Notes were completed sooner, which meant that downstream clinicians were sometimes no longer waiting for a consult note before acting, and patients arriving in the emergency department could have a completed primary care note from earlier that day. The tool could preserve patient voice, produce coherent histories of present illness, and generate summaries that clinicians themselves sometimes described as useful.

Traverse succeeded at the problem clinicians had come to experience most vividly: the cost of producing a note. It sharply reduced the effort required to generate a draft that clinicians could sign and that was treated as sufficient for prevailing billing, review, and circulation requirements. But success on that dimension did not settle whether the note remained useful as a resource for subsequent clinical work. Making it useful still required deciding what mattered, organizing the clinical story, marking uncertainty, and anticipating what another reader would need to know.

This distinction became visible in the way clinicians used the tool. Before Traverse, even templated or copied authorship of notes required the clinician to construct a representation of the patient. They still had to decide what to include, what to remove, and how to render the patient's situation for future use. In Berg's terms, writing remained one practice through which clinical attention and judgment were organized (Berg, 1996; Berg and Bowker, 1997). Traverse changed the starting point. The clinician no longer began with the problem of composing an account of the patient's case but with a generated account, and decided whether it was acceptable enough to sign.

Ratification was not a neutral substitute for authorship. Authorship asks, "What needs to be said?" Ratification changed the final authorial test from generative selection to error detection. The first asks what the record must carry forward; the second asks whether the generated account contains a sufficiently salient reason not to sign it. A physician who reviewed an AI-generated note immediately after a visit still held the encounter in memory - they at least mostly remembered the patient, the conversation, and the plan. Against that memory, the note did not have to carry the patient's situation on its own; it only had to avoid contradicting what the physician already knew. One provider after glancing quickly at a note said "it's complete. I know what's there is correct." Her memory allowed her to make sure that nothing was wrong; the downstream reader would not have that scaffold.

A striking example of a salient commission error came from a primary care provider whose patient had recently had brain surgery. In the exam room the patient had complained of numbness on her scalp surrounding the incision. Upon a quick skim of the Traverse-generated visit summary, the physician noted that the text said "Patient self describes as a numbskull." This line was, according to the provider, hilarious. She got a good chuckle with her colleagues, and was able to delete the line and correct the note. A similar thing happened when an AI generated note said that a patient had bears in his trashcans. The provider noticed and commented to me that *he* had bears in his trashcans, not the patient. These situations stood out enough to be noticed.

More difficult to detect were plausible inaccuracies and omissions that did not announce themselves. On more than one occasion I witnessed doctors skim an AI-generated record and conclude that it was "good enough to sign," despite errors involving the side of the body of the complaint, the number of weeks since a focal event, the medication dose in mg as opposed to mL, or a statement that the wife was not in the room when she in fact was. I noticed these discrepancies only because I had been in the room and had recorded the encounter in my own fieldnotes. A review criterion of "not obviously wrong" could detect errors that jarred against fresh memory, but did little to reveal plausible inaccuracies or missing information. Such notes could pass the immediate test of ratification without establishing either their adequacy as accounts of prior work or their usefulness to a future clinician. Ratification changed the test applied at the point of production. Authorship began by asking what the note needed to contain; ratification began with a generated account and asked whether it contained a sufficiently visible reason not to sign it. The first test required affirmative selection and synthesis. The second privileged the detection of salient defects.

When the mistake was not so egregious the pattern of use was easy to see. The cadence of review was often a brief scan accompanied by affirmations that the note seemed acceptable: "yep, yep," "that makes sense." In one inpatient observation, a clinician left briefly while Traverse continued recording the discussion of a particular patient, telling the PA to "keep talking" because Traverse was listening. When he returned, he did not ask the PA to repeat what had been said. After generating the note, he scanned it briefly, and said "I don't see anything funky. I'm sure I may have missed something but I don't care because the cost of taking the time to fix it isn't worth it." When scanning the note, he noticed generated material he felt was redundant. He said "Maybe I should delete this extra info, but actually, I'll just leave it because it generated it and why take the time to delete it." The note was good enough to sign. This directly decreased the cost of producing the record for him, but increased the search cost for a downstream clinical reader.

One primary care provider captured the distinction between the search for something "funky" and pure omission when Traverse had failed to include a discussion about an at-home blood pressure cuff as related to the patient's hypertension. As he added it into the note, he said to me "this is something I know that people are not doing... making sure they're complete. They're only checking to make sure nothing is wrong." His comment clarified the weakness of ratification as a safeguard. A statement that was plainly wrong could jar against memory and invite correction. Missing information by contrast did not announce itself. A note could pass review because nothing in it appeared false while still failing to contain what a future reader would need to treat the patient. Here, completeness did not mean that every required field was populated. It meant that the note contained the information a later clinician would need to understand and act on the hypertension plan.

Traverse also sometimes changed what clinicians oriented to during clinical work itself. In an ICU observation early in the inpatient rollout, an attending who had just begun using Traverse interrupted an Advanced Practitioner's (APP) patient presentation. Typically, the APP would report labs and clinical developments and the attending would discuss them. This time, as the APP began to report labs, the attending cut him off and said, "I mean, I can see it." He then read laboratory values aloud so that Traverse would capture them. Later, he reopened an echocardiogram and read the impression verbatim into the recording, saying they should mention it to the AI to make sure it was captured. In such moments, rather than documenting clinical work that had occurred, it began to reorganize what clinicians oriented to during the work. Both what needed to be discussed and what needed to be heard by the tool.

The behavioral shift was not uniform, and some clinicians perceived the limitation. One leader involved in piloting Traverse stopped using it because he found the notes too generic and too long. Yet because he was part of the group responsible for adopting the tool, he felt he could not simply opt out and resumed use because he felt he was "supposed to." His experience revealed the limits of defining success through adoption or burden relief: weaknesses in the notes' clinical usefulness remained visible, but Traverse had become organizationally legitimate. Once the tool became the system's solution to documentation burden, even skepticism among its champions could be folded back into implementation.

For many clinicians, however, the dominant experience remained relief rather than compromise. One clinician described spending substantial time early on combining Traverse-generated sections, smoothing paragraphs and fonts, and making notes "look pretty." Over time, she concluded that the notes were good enough to bill on and stopped investing that effort. Seeing that one of her providers was still charting at 2 a.m. post Traverse, she said she needed to have a "come to Jesus" and reassure him that it was acceptable if hypertension appeared twice in the note: "Don't kill yourself to make the note fine." This was not indifference to care. It was the logic of burden relief. Once Traverse produced a note that was adequate enough as an account under prevailing evaluative standards, the additional work required to make it concise and useful for ongoing clinical work remained part of the very burden the tool had promised to remove. The use cost of redundancy had become a delayed cost borne by a future reader. Because that cost did not enter the relieved author's immediate decision about whether to edit, it exerted little pressure on production.

Traverse made visible a distinction that had been emerging throughout the fieldwork: the note's *account adequacy* and its *clinical usefulness* were no longer moving together. Authorship had never guaranteed that the two dimensions would align; human-written notes could also be bloated, inaccurate, or unhelpful. But authorship had remained one practice linking them. Composing the note required clinicians to select and synthesize what mattered, and visible evidence of that work gave later readers some reason to expect that the note might be worth using. Traverse preserved the production of text while weakening this coupling mechanism. A long note no longer reliably indicated that a clinician had spent time doing that work. Before Traverse, providers sometimes treated the effort embodied in a "super long note" as evidence that the documented need was real, even when insurers denied the claim. After Traverse, text became abundant, but its value as a signal of authorial effort weakened. This pattern extends beyond Yankee Health: a recent synthesis of ambient-scribe studies found that note length increased across settings while the proportion written by clinicians declined substantially (Kanaparthy et al., 2025). The review similarly warned that longer AI-generated notes may add redundancy without increasing clinical relevance.

When one clinician encountered a particularly strange note, he asked, "Was this written by AI? Is this made up?" Another described his own AI-written notes as difficult to read because "everything is in the same font and redundant and it's hard to find what I need," in almost the same breath as claiming that the output was "decent." AI notes were not uniformly poor; length, completeness, and polish no longer reliably signaled the kind of expert synthesis that would let a provider quickly know "what's up" with a patient.

The tool also reproduced the multiple definitions of "good" described earlier. One primary care provider pointed me to where Traverse had captured a conversation about the patient's yellow ottoman. "Patients love that I have their ottoman included. Everything is all about customer service, but they're not customers." He observed that patients feel it signals his attention to the small things about their life and could matter for his patient satisfaction scores and eventual payment. But he also lamented the search cost of dealing with this non-clinical information. Traverse could preserve patient voice and relational detail, which were not valueless, of course, but the same abundance of captured text could bury the clinical information a future clinician reader might need. Traverse could therefore produce a complete and personable account without making the note economical for a future clinician to use.

The result was a new dissonance. Clinicians celebrated Traverse's relief from documentation burden while criticizing AI-generated or AI-like notes as redundant, bloated, generic, or hard to use. One clinician complained, "We either have notes that say nothing or notes that say too much. I'm never going to be able to read all of this." Another added "no AI statements" to a note "just to make it readable." Yet when clinicians reviewed notes they had just generated, with the encounter still fresh in memory, they sometimes described them as useful summaries. One called a Traverse note a "really good summary" and "a really nice read," saying it would serve whoever needed to read it. Fresh memory supplied context that a downstream reader would have to recover from the note itself. A note could therefore appear clinically useful to its producer while imposing use costs on a later reader. Traverse distributed these effects asymmetrically: relief was immediate, vivid, and borne by the author, whereas any loss of clinical usefulness was delayed, diffuse, and usually encountered through someone else's note.

These observations do not establish an average decline in note content, that is a separate empirical question. What I observed directly was a change in behavior. Traverse shifted clinicians from composing notes to ratifying generated drafts, making the work of selecting what mattered, synthesizing the case, and writing for future readers less routine. The change was visible within the same clinicians and settings through reduced review time, less editing, and weaker knowledge of note content. One primary care provider illustrated both sides of this shift. Before Traverse, she conducted highly organized pre-charting, including careful review of other encounter notes, and often remained on her computer past midnight finishing her own documentation. She described severe burnout and called the EMR the "bane of my existence." After Traverse, she reviewed little beyond laboratory results and imaging, explaining that her "encyclopedic memory" and Traverse meant she no longer needed to review prior notes. At 4:53 p.m. one day, she said, "If Odyssey decides to open, I'll be out by five." Across roughly a dozen post-lunch visits, she spent no more than 37 seconds reviewing any generated note and signed many without looking. Traverse thus reduced production burden while also reducing *clinician reliance on the note* and *clinician authorship investment*, the two behavioral relationships underlying recursive withdrawal.

AI scribing exposed and intensified, rather than created, *differential coupling*. Traverse entered a record system already shaped by asymmetric correction and recursive withdrawal. In the model, it altered several links without changing the underlying feedback structure. It sharply reduced the production-cost component of *documentation burden* by making it easier to generate a note that could be signed and was treated as adequate under prevailing evaluative standards. This was genuine organizational success: notes were completed sooner, clinicians experienced immediate relief, and documentation remained available for billing, audit, and review.

At the same time, shifting clinicians from authorship to ratification reduced *clinician authorship investment* and weakened the effort signal through which downstream readers inferred that another clinician had selected and weighed what mattered. The benefit to the producer was immediate and visible; any use costs appeared later through search, verification, and reconstruction by downstream readers. My evidence establishes this redistribution of effort and attention, not that downstream use costs exceeded the production costs Traverse removed.

Traverse was therefore relief without repair. It lowered the effort required to satisfy prevailing standards of *account adequacy* without strengthening *use-based note repair* or rebuilding the authorship–readership relationship through which *clinical usefulness* had been sustained. It did not eliminate the work required to produce a clinically useful note; it repositioned selection and synthesis as discretionary editing that the technology made easier to forgo.

None of this required carelessness or indifference. Clinicians and the organization responded reasonably to the costs and feedback they faced. The value lay in clinical authorship that remained entangled with a burdensome documentation process. By relieving that process without preserving the selection and synthesis authorship supplied, automation could remove value along with burden.

5 Discussion

TThis study distinguishes two dimensions of a shared record that are often treated as one: *account adequacy* and clinical usefulness. Account adequacy concerns whether the record satisfies the standards applied to it as an account of prior work; clinical usefulness concerns whether it carries the judgment needed for ongoing work. The two dimensions can reinforce one another, but they need not move together. At Yankee Health, clinical notes continued to be completed, signed, billed upon, audited, and used to authorize further action even as clinicians increasingly treated them as poor guides to what had happened and what to do next. The record did not disappear. What weakened was the authorship–readership relationship through which professional judgment became portable.

I explain this divergence through two feedback processes. Through *asymmetric correction*, failures to meet formal standards returned quickly to identifiable authors as queries, correction requests, training, workflow constraints, or system redesign. Through *recursive withdrawal*, failures to meet clinicians' needs were repaired locally by downstream readers through search, verification, side communication, memory, and rework. These repairs preserved immediate care, but they rarely changed the record. Instead, they reduced reliance on notes, lowered anticipated readership, and made careful authorship less rewarding. Together these processes produced *differential coupling*: the note became more tightly connected to accountability requirements and less tightly connected to the clinical usefulness it was also expected to support.

The arrival of the AI scribe did not create these feedback structures. It altered how much human authorship was still required within them. Traverse relieved a genuine burden by making notes that satisfied prevailing evaluative standards cheaper to produce. But because years of asymmetric correction had already shaped the record toward requirements whose failures returned, the technology preserved those properties more readily than the selection and synthesis through which clinicians carried judgment forward. The shift from authorship to ratification therefore served as a revealing shock. It exposed a *burden-value entanglement*: the value lay not in the burden itself, but in clinical authorship that remained embedded in the burdensome process.

5.1 Documentation as organizational infrastructure

The first contribution is to theorize documentation as organizational infrastructure rather than as trace data, compliance output, bureaucratic residue, or a technology-use outcome. Written records make organizational knowledge durable, portable, and available beyond particular officeholders, allowing organizations to coordinate, evaluate, and govern work that leaders cannot continuously observe (Stinchcombe, 2001; Weber, 1978; Yates and Chandler, 1951). This study shows that such portability is not produced by storage alone, but depends on a reciprocal relation between authors and readers: workers invest judgment in records partly because they expect others to read and use them, and readers rely on records partly because they believe authors invested such judgment.

This dynamic identifies a paradoxical consequence of formalization. The classic bureaucratic accomplishment of records is to make coordination less dependent on particular people. At Yankee Health, however, notes could remain adequate under prevailing evaluative standards while becoming less useful for ongoing work. Clinicians recovered missing context through calls, messages, relationships, repeated work, and memory. The hospital thus became more thoroughly documented while becoming more dependent on particular people to know what had happened and what should happen next. In the language of organizational knowledge strategy, personalization did not emerge as a deliberate alternative to codification (Hansen et al., 1999); it emerged as local compensation for a codified system that continued to function on one dimension while weakening on another.

This contribution is bounded. The mechanism should be strongest in mandatory shared record systems with three conditions in particular. First, future users of the record must also be future producers of similar records, linking current reliance to anticipated future readership. Second, the record must have both representational value, in that it documents work for others and also constitutive value, in that producing it helps workers to select, synthesize, remember, or communicate what matters. Third, shortfalls in account adequacy must have institutionalized return paths, while failures of clinical usefulness are delayed, distributed across downstream users, and susceptible to local compensation. In such systems, the organization may see formal production and compliance while being blind to erosion in the record's use value.

5.2 Differential coupling and unequal organizational learning

The second contribution extends theories of decoupling by identifying *differential coupling* within a single organizational artifact. Classic decoupling separates formal structure from technical activity (Meyer and Rowan, 1977); selective coupling describes how organizations combine elements of competing logics (Pache and Santos, 2013). Differential coupling instead describes how different dimensions of the same object become governed by feedback systems that differ in strength, speed, and enforceability. The note was not simply coupled or decoupled from work. It became more tightly coupled to account adequacy and more weakly coupled to clinical usefulness.

The mechanism was unequal organizational learning. Evaluative shortfalls were visible, attributable, and returnable: they generated queries, training, complaints, or system changes. Failures in use were more often absorbed through search, verification, side communication, memory, and rework. This pattern is consistent with Ouchi's argument that organizations rely more heavily on formal control where performance can be specified and verified (Ouchi, 1979). My findings sharpen that insight: the consequence was that harder to measure failures in use were repaired locally in ways that prevented them from generating feedback capable of changing future record production. This extends research on workarounds and capability traps (Repenning and Sterman, 2002; Tucker and Edmondson, 2003; Senge, 1990): local compensation reduced reliance on the note, anticipated readership, and ultimately authorship investment, thereby changing the conditions of future production.

5.3 AI and the burden-value entanglement

The third contribution concerns a problem that becomes especially visible under automation: burdensome work may contain activities whose value is easy to miss precisely because they are embedded in the burden. The implication is that reducing effort requires identifying which parts of a practice are costly and which perform otherwise hidden organizational functions. This is especially important in expert work, where valuable judgment is often situated, tacit, and difficult to separate cleanly from the activity through which it is expressed (Polanyi, 1966; Nelson and Winter, 1982).

AI makes this problem consequential because it can preserve an organizational artifact while changing the process through which that artifact acquired value. A record may remain complete, legible, and formally adequate even when automation alters the situated acts of selection, interpretation, and synthesis that previously made expert judgment portable. Evaluating the technology only by the persistence or apparent quality of the artifact can therefore obscure what the artifact is no longer able to do for subsequent users.

I call this a *burden-value entanglement*: costly and value-producing activities are bundled within the same practice, so removing the former can inadvertently remove the latter unless its function is deliberately reconstructed elsewhere. This extends research on AI and expert work by shifting attention from whether technology substitutes for human labor to which organizational functions are bundled inside the labor being automated. It also qualifies a simple efficiency logic: reducing the cost of producing an output need not preserve every dimension of the output's value. The relevant design question is therefore not only which burdens AI can remove, but which forms of judgment, coordination, or learning must be deliberately preserved when it does (Elish, 2019).

5.4 Practical implications

The practical implication is not to reject AI scribes or to preserve documentation burden, but to distinguish wasteful effort from work that performs a valuable organizational function. Automation should therefore be accompanied by mechanisms that preserve what burden reduction might otherwise displace. Most importantly, organizations should create return paths from downstream users to upstream producers, require clinicians to contribute an explicit synthesis of what matters and remains uncertain, and design review around omissions as well as visible errors. These changes would make *clinical usefulness* more observable and less dependent on discretionary effort.

Organizations should also evaluate documentation technologies on more than production efficiency and compliance. Completion time, clinician satisfaction, billing adequacy, and audit performance capture only part of the record's value. Measures of downstream use such as readability, recoverable reasoning, repeated searching or testing, handoff quality, and reliance on side channels are needed to detect failures that are otherwise absorbed locally. Workarounds are especially informative in this respect: rather than treating them only as resistance or noncompliance, leaders can use them as evidence of functions the formal system no longer performs reliably (Tucker and Edmondson, 2003; Repenning and Sterman, 2002).

Finally, seemingly simple architectural fixes create new governance problems. Separate clinical and administrative records, or AI-generated summaries layered over a full record, may reduce burden or search cost but also create disagreement over which representation is authoritative. Responsibility is unlikely to disappear simply because production or summarization is automated; professionals may remain accountable for outputs they no longer fully produce or inspect (Elish, 2019). The managerial challenge is to ensure that the functions displaced by automation remain visible, assigned, and supported somewhere in the system.

This is especially important because workarounds often look, from above, like loopholes. One informant asked me to promise that Yankee Health would not read this paper as a catalogue of loopholes to close. Her candid request captures the managerial implication: informal adaptations are evidence of work those systems no longer perform reliably. Closing the workaround without restoring that function may produce a more compliant organization but a less capable one.

5.5 Future research

Future research should examine three consequences that this study identifies but cannot fully assess: measurement, cognition, and cross-setting variation. First, future work should develop better measures of whether records remain useful in practice. Completion and availability reveal little about downstream dependence. More informative indicators may include repeated searching, side-channel communication, rework, handoff failures, or other traces of effort required to compensate for the record. Such measures could make declining usefulness visible before it appears only as frontline frustration.

A second question concerns the cognitive consequences of shifting from authorship to ratification. If writing is part of how experts select, remember, and integrate information, then delegating composition may change not only the artifact but also the cognition that surrounds it. Future work should test whether ratification affects recall, learning, diagnostic reasoning, or continuity over longer time horizons.

Finally, the effects of automation should vary with the feedback structure into which it is introduced. AI may complement expert authorship where downstream use remains visible and valued, but substitute for it where evaluation is concentrated on other dimensions of the record. Comparative research across documentary systems, from engineering logs and incident reports to customer records and performance reviews, could identify when automation preserves multiple organizational functions and when it produces differential coupling.

6 Conclusion

The clinical note began as a record a clinician wrote for the next clinician. It became a record written for everyone except, increasingly, the next clinician. The contradiction with which this paper opened, that physicians experience the note as both indispensable and unbearable, resolves into something less paradoxical and more troubling: the note became unbearable through the same process that made it seem dispensable.

When technology offered to lift the burden, clinicians dropped it. The decision was reasonable. That is why the outcome was hard to see and will be hard to reverse. AI scribing made the failure visible because it relieved a real burden. But the act relieved was also one of the acts through which clinical meaning entered the record. When burden and value are carried by the same act, organizations must do more than remove the burden. They must rebuild the system through which value is produced.

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